VPM-5.3: Verifier-Native Compressive Predictive Mechanisms

A support-preserving architecture with a language-normalizing neural substrate for certified inference, compression, tool use, and continual mechanism formation

Architecture draft

May 2026

Abstract

VPM-5.3 is a budgeted posterior machine over typed executable mechanisms, a verifier-native neural substrate, proof-carrying compressed libraries, and an explicit natural-language contract normalizer. The revision treats the principal failure modes as first-class state: verifier dependence, support loss, substrate omission, ambiguous context, semantic collapse, source and rebuttal miss, realization drift, scalar-risk cancellation, active-memory bloat, non-reusable residual entropy, and training compute spent away from the certificate boundary. The substrate remains an amortizer rather than an authority: it proposes typed event graphs, predictive-state tests, mechanisms, atom plans, source/rebuttal sets, repairs, and renderings, while certification is produced only by split-clean independent evidence, calibrated recall bounds, executable replay, authority proofs, and vector safety gates.

The system is a controlled rate-distortion architecture rather than a generic text model. It can compress only reusable, contract-stable structure whose macro has an expansion witness, positive sequence-valid held-out gain, bounded branch increase, bounded context/semantic/source/rebuttal/realization error, and acceptable capability risk; incompressible residuals remain archival or lossy sketches with explicit distortion. Intelligence is identified with certified frontier movement: reducing the support gap by adding primitives, tests, memories, verifiers, tools, causal interventions, semantic formalisms, and compression capsules that lower held-out certified cost without increasing false-pass, miss, or risk budgets. Safety is vector-valued and transcript-level: impact, privacy, exfiltration, capability, influence, conflict, model, and dependency risks are accumulated by confidence sequences and no scalar average can cancel a violated component. Training is execution-first and allocation-controlled: exact search, verification, retrieval, red-team, replay, and neural updates are scheduled by marginal certified utility under dual prices for scarce labels, calibration data, verifier calls, and exact execution. When p_v^+ is too large, verifier dependence is unresolved, $\eta_v = 0$, support, context, semantic, source, rebuttal, realization, substrate, dependency, compression-gain, or vector-risk bounds fail, or intent entropy remains high, the system decomposes, grounds, asks, narrows, archives, or abstains rather than claiming compression, safety, or intelligence.

1 Contract, ledger, and non-negotiable invariants

A task is

$$T = (O_{0:t}, A, B, R, H, \mathcal{F}_t), \tag{1}$$

where $O_{0:t}$ are observations, A admissible actions, B resource budgets, R risk policy, H scoped history, and $\mathcal{F}_t$ the filtration containing every observation, proposal, verifier choice, tool call, and human bit. A contract is

$$\begin{aligned}\Gamma = (&Y, S, V, P, L, F, Q, \Pi_{\text{perm}}, \Pi_{\text{priv}}, \Pi_{\text{audit}}, \Pi_{\text{split}}, \Pi_{\text{risk}}, \Pi_{\text{dom}}, \Pi_{\text{sub}}, \Pi_{\text{sem}}, \Pi_{\text{src}}, \Pi_{\text{ctx}},\\ &\Pi_{\text{rebut}}, \Pi_{\text{real}}, \Pi_{\text{style}}, \Pi_{\text{infl}}, \Pi_{\text{dep}}, \Pi_{\text{cap}}, \Pi_{\text{compute}}).\end{aligned} \tag{2}$$

where Y is the output type, S the success predicate, V the evidence threshold, P provenance requirements, L latency/resource limits, F failure behavior, Q rendering constraints, Π_{perm} permission policy, Π_{priv} privacy/declassification policy, Π_{audit} transcript requirements, Π_{split} certification-split policy, Π_{risk} vector impact/privacy/capability/influence policy, Π_{dom} domain-admissibility policy, Π_{sub} neural-substrate support policy, Π_{sem} semantic-contract policy, Π_{src} source/evidence-recall policy, Π_{ctx} context/reference policy, Π_{rebut} contradiction-search policy, Π_{real} atom-to-text realization policy, Π_{style} nonclaim style policy, Π_{infl} influence/manipulation policy, Π_{dep} verifier/retriever/model-dependence policy, Π_{cap} compression-capacity and rate-distortion policy, and Π_{compute} dual compute-allocation policy. Every semantic atom a_j, rendered span y_j, action a, memory write m, and declassification d is entered in a ledger

$$\begin{aligned}\Lambda = (&id, type, mode, scope, ctx, sem, parents, entail, src, rebut, real, auth, prov, split, gen, dep, shift, verifiers,\\ &cert, cost, \mathbf{r}, taint, frontier, time, hash).\end{aligned} \tag{3}$$

Unledgered state may rank proposals but cannot be rendered as fact or used to authorize side effects.
A contract is admissible only when the evidence economy is favorable. Define the online certifiability vector

$$\Xi_\Gamma(T) = \begin{pmatrix} \hat{c}_v/(\widehat{U}_\Gamma + \epsilon),\ \hat{p}_v^+,\ \hat{\eta}_v,\ \hat{\epsilon}_{dep},\ \hat{d}_{shift},\ \hat{\epsilon}_{support},\ \hat{\rho}_{reuse},\ \hat{b}_{mech} \\ \hat{H}_\Gamma,\ \hat{\epsilon}_{sub},\ \hat{\epsilon}_{sem},\ \hat{\epsilon}_{src},\ \hat{\epsilon}_{ctx},\ \hat{\epsilon}_{rebut},\ \hat{\epsilon}_{real},\ \widehat{\Delta\,\text{Front}},\ \hat{\mathbf{r}}_{cum} \end{pmatrix}, \tag{4}$$

where $\hat{c}_v$ is terminal verification cost, $\hat{p}_v^+$ the usable false-pass upper bound, $\hat{\eta}_v$ independent evidence weight, $\hat{\epsilon}_{dep}$ un-residualized dependence among verifiers, retrievers, generators, and learned checkers, $\hat{d}_{shift}$ calibrated distribution-shift distance, $\hat{\epsilon}_{support}$ lost-support bound, $\hat{\rho}_{reuse}$ expected mechanism reuse, $\hat{b}_{mech}$ residual branching after known macros, $\hat{H}_\Gamma$ posterior entropy of user intent/contract, $\hat{\epsilon}_{sub}$ probability that the neural substrate omits all certifying traces, $\hat{\epsilon}_{sem}$ semantic-contractization loss, $\hat{\epsilon}_{src}$ source/evidence-recall loss for factual atoms, $\hat{\epsilon}_{ctx}$ context/reference/vagueness loss, $\hat{\epsilon}_{rebut}$ material contradiction-recall loss, $\hat{\epsilon}_{real}$ atom-to-rendered-text round-trip loss, $\widehat{\Delta\,\mathrm{Front}}$ the lower confidence movement of the compression/intelligence frontier, and $\widehat{\mathbf{r}}_{cum} \in \mathbb{R}_+^{J_R}$ cumulative residual risk by channel. The hard domain predicate is

$$\begin{aligned}\mathrm{DomOK}(T,\Gamma) = \mathbf{1}\{&\hat{c}_v \le \kappa_c(\widehat{U}_\Gamma + \epsilon) \wedge \hat{p}_v^+ \le p_{max} \wedge \hat{\eta}_v \ge \eta_{min} \wedge \hat{\epsilon}_{dep} \le \epsilon_{dep,max} \\ &\wedge \hat{d}_{shift} \le d_{shift,max} \wedge \hat{\epsilon}_{support} \le \epsilon_\Gamma \wedge \hat{\rho}_{reuse} \ge \rho_{min} \wedge \hat{b}_{mech} \le b_{max} \\ &\wedge \hat{\epsilon}_{sub} \le \epsilon_{sub,max} \wedge \hat{\epsilon}_{sem} \le \epsilon_{sem,max} \wedge \hat{\epsilon}_{src} \le \epsilon_{src,max} \wedge \hat{\epsilon}_{ctx} \le \epsilon_{ctx,max} \\ &\wedge \hat{\epsilon}_{rebut} \le \epsilon_{rebut,max} \wedge \hat{\epsilon}_{real} \le \epsilon_{real,max} \wedge \widehat{\Delta\,\mathrm{Front}} \ge -\epsilon_{front} \wedge \widehat{\mathbf{r}}_{cum} \preceq \mathbf{r}_{max}\}, \qquad \mathrm{IntentOK}(T,\Gamma) = \mathbf{1}\{\hat{H}_\Gamma(T) \le h_\Gamma\}.\end{aligned} \tag{5}$$

Routing is part of inference rather than a post-hoc caveat. The cases are priority-ordered, so hard certificate failure cannot be masked by grounding or decomposition:

$$\mathrm{Route}(T,\Gamma) = \begin{cases} \text{Abstain}, & \hat{p}_v^+ = 1 \vee \hat{\eta}_v = 0 \vee \hat{\epsilon}_{dep} = 1 \vee \hat{c}_v > \kappa_c(\widehat{U}_\Gamma + \epsilon) \vee \hat{\epsilon}_{sem} = 1 \vee \hat{\epsilon}_{ctx} = 1 \vee \hat{\epsilon}_{real} = 1, \\ \text{Ask}, & (\mathrm{IntentOK} = 0 \vee \hat{\epsilon}_{sem} > \epsilon_{sem,max} \vee \hat{\epsilon}_{ctx} > \epsilon_{ctx,max}) \wedge C_{ask} \le B_{ask}, \\ \text{narrow}, & \hat{\epsilon}_{support} > \epsilon_\Gamma \vee \hat{\epsilon}_{sub} > \epsilon_{sub,max} \vee \hat{\epsilon}_{dep} > \epsilon_{dep,max} \vee \hat{\epsilon}_{real} > \epsilon_{real,max} \vee \neg(\hat{\mathbf{r}}_{cum} \preceq \mathbf{r}_{max}), \\ \text{ground}, & (\hat{\epsilon}_{src} > \epsilon_{src,max} \vee \hat{\epsilon}_{rebut} > \epsilon_{rebut,max}) \wedge C_{src} \le B_{src}, \\ \text{solve}, & \mathrm{DomOK} = 1 \wedge \mathrm{IntentOK} = 1, \\ \text{decompose}, & \exists\{T_i,\Gamma_i\}_{i\le k} : \prod_i \mathrm{DomOK}(T_i,\Gamma_i) = 1 \wedge C_{glue} \le B_{glue}, \\ \text{archive}, & \hat{\rho}_{reuse} < \rho_{min} \wedge \widehat{\Delta\,\mathrm{Front}} \le 0 \wedge \widehat{U}_\Gamma \le C_{cert}, \\ \text{Abstain}, & \text{otherwise.} \end{cases} \tag{6}$$

The primary operating region is therefore $\mathcal{P}_{cert} = \{\mathcal{P} : \Pr_{T\sim\mathcal{P}}(\mathrm{Route}(T,\Gamma) \in \{\text{solve}, \text{decompose}, \text{ground}\}) \text{ is high}\}$; broad language, aesthetics, social judgment, and novelty remain admissible only as a typed mixture of certified atoms, scoped assumptions, explicitly marked uncertainty, questions, refusals, or nonclaim style.

Proposition 1 (Verifier, support, and meaning collapse)**.** *If all terminal verifiers for a claim have $\hat{\eta}_v = 0$ or $\hat{p}_v^+ = 1$, or if unresolved dependence has $\hat{\epsilon}_{dep} = 1$, or if $\hat{\epsilon}_{support} = 1$, $\hat{\epsilon}_{sub} = 1$, $\hat{\epsilon}_{sem} = 1$, $\hat{\epsilon}_{ctx} = 1$, $\hat{\epsilon}_{real} = 1$, or $\hat{\epsilon}_{src} = 1$ for a factual atom requiring external grounding, or $\hat{\epsilon}_{rebut} = 1$ for an open-world atom requiring contradiction search, or if any mandatory risk coordinate has exhausted its budget, then no positive scoped certificate can be obtained for that atom under the gate.*

Proof. For $\hat{\eta}_v = 0$, $E_{v,r} = 1$. For $\hat{p}_v^+ = 1$ and a pass, $E_{v,r} \le 1$; a failure also gives $E_{v,r} \le 1$. Hence M_r cannot exceed its family-spending threshold. Unit support, dependency, substrate omission, semantic loss, context loss, source-recall loss, rebuttal-recall loss, or realization loss sets the corresponding cap $[-\log(1+\epsilon_0)]_+ = 0$; an exhausted mandatory risk coordinate sets $\mathrm{Cert}_{risk} = 0$. The scoped certificate is a minimum over observation, entailment, source, rebuttal, support, substrate, context, semantic, realization, domain, privacy, and authority caps, and the gate is conjunctive. □

Certified utility is measured on the whole transcript, not merely on final text:

$$\begin{aligned}S_{\mathrm{ver}}(f;\mathcal{P},B) = \mathbb{E}_{T\sim\mathcal{P}}\Big[&U_\Gamma(T,f)\mathbf{1}\{\mathrm{Cert}_T(f) \ge V_\Gamma\}\mathbf{1}\{\mathrm{Gate}(f,T,\Gamma) = 1\} \\ &- \lambda_F F - \lambda_L L - \lambda_M|\mathcal{A}_{act}| - \lambda_H H_{bits} - \lambda_C C_{hidden} - \lambda_R \mathrm{Risk} - \lambda_U U_{unsupported}\Big].\end{aligned} \tag{7}$$

A comparison is valid only under matched training data, wall-clock, FLOPs, memory growth, tool authority, search budget, verifier budget, human bits, and transcript logging. Unreported test-time search, external inference, or latent human feedback sets $C_{hidden} = \infty$ for the comparison.

Invariant 1 (Proposal, evidence, authority, and compression are separate)**.** *Retrieved text, popularity, neural confidence, tool output, imitation traces, macro frequency, and search success may propose hypotheses. They cannot raise* Cert*, grant authority, enter active memory, or declassify data except through the verifier, gate, and compression rules below.*

Invariant 2 (Claim-family validity)**.** *A certificate is not a scalar confidence score. For every adaptive family of candidate claims, actions, memory writes, and releases, the ledger stores a spending measure α_j with $\sum_j \alpha_j \le \alpha_\Gamma$. Rendering or action thresholds are applied to excess log-evidence after this family charge. A mined claim whose selection used the audit split has certificate weight zero until replayed on a disjoint split.*

Invariant 3 (Support conservation)**.** *A pruning, canonicalization, retrieval, sparse-attention, semantic parser, source-ranking, rebuttal-ranking, renderer, or retrieval-ranking step may discard alternatives only with a ledgered, recall-calibrated upper bound on lost posterior mass, context commitments, source coverage, material contradictions, semantic interpretations, atom realizations, or certified utility. Otherwise it is a proposal heuristic; the final answer must ask, abstain, or re-run an exact bounded stage.*

Invariant 4 (Semantic faithfulness). *A generated sentence is not evidence about its own meaning. Every factual, mathematical, executable, private, or action-guiding span is first reduced to semantic atoms and then checked by scoped entailment from ledgered parents. Style, empathy, refusal, and creative spans may be uncertified only when the renderer proves that they contain no gated atom and inherit no private or capability-bearing taint.*

Invariant 5 (Certifiability before ambition). *An answer is active only on a certified or decomposable contract. If cheap independent tests do not exist, user intent is underdetermined, context or vague predicates are unresolved, semantic interpretations are not separated, source or rebuttal recall is unbounded, atom-to-text realization is unbounded, verifier cost exceeds expected utility, support recall is unbounded, cumulative risk is exhausted, or the base substrate cannot recall a certifying trace family, the architecture must reduce the task, ground, ask, narrow, or abstain; dense language fluency is not evidence.*

Invariant 6 (Vector risk and dependence before aggregation). *Risk and evidence are never averaged before gating. Let $\mathcal{J}_R$ index impact, privacy, exfiltration, capability, influence, conflict, model, and dependence risk. Every claim, action, memory write, release, macro admission, retrieval step, and renderer span carries $\mathbf{r} \in \mathbb{R}_+^{J_R}$ and a dependence class* Dep. *A scalar utility may rank proposals only after the componentwise inequalities $\mathbf{r} \preceq \mathbf{B}_R$ and the dependence policy Π_{dep} hold; otherwise the state is narrowed, split, residualized, or abstained.*

2 Typed mechanisms and calibrated evidence

Definition 1 (Mechanism). *A mechanism is*

$$\mu = (\mathcal{S}, \mathcal{T}, \mathcal{O}, \Pi, U, \chi, \nu, \alpha, \ell, \omega, r, \lambda), \tag{8}$$

where $\mathcal{S}$ is typed latent state, $\mathcal{T}$ transition, $\mathcal{O}$ observation/render map, Π executable procedure or policy, U contract utility, χ invariants, ν verifier obligations, α authority requirements, ℓ code length, ω scope/fragility metadata, r replay statistics, and λ data/authority taint. Equivalence is contractual, counterfactual, and trace-preserving:

$$\mu_i \sim_\Gamma \mu_j \iff \begin{cases} \forall q \in Q_\Gamma : \text{Exec}(\mu_i, q) = \text{Exec}(\mu_j, q), \\ \forall a \in A_\Gamma^{cf} : \text{Exec}(\text{do}(a, \mu_i), q) = \text{Exec}(\text{do}(a, \mu_j), q), \\ \Lambda(\text{Exec}(\mu_i, q)) \equiv_\Gamma \Lambda(\text{Exec}(\mu_j, q)). \end{cases} \tag{9}$$

Posterior mass is assigned to $[\mu]_\Gamma$, not to syntax.

The recurrent state is

$$Z_t = (C_t, M_t, S_t, Q_t, V_t, P_t, A_t, G_t, K_t, D_t, E_t^{sem}, C_t^{ctx}, R_t^{src}, R_t^{rebut}, \Omega_t^{real}, h_t), \tag{10}$$

where C is a scene/contract posterior, M a mechanism posterior, S predictive state, Q active tests, V verifier records, P provenance graph, A authority labels, G e-graph/canonicalization state, K support-loss certificates, D discourse state, E^{sem} semantic atoms and entailment obligations, C^{ctx} context/reference commitments, R^{src} source/evidence coverage state, R^{rebut} material-rebuttal coverage state, Ω^{real} atom-to-rendering obligations, and h compact neural state.
The active dictionary is

$$\mathcal{D}_t = \{d_k = (\tau_k^{in}, \tau_k^{out}, f_k, \chi_k, \nu_k, p_k, \Omega_k, r_k, \lambda_k, \sigma_k^{sem}, \rho_k^{src}, \rho_k^{ctx}, \rho_k^{rebut}, \rho_k^{real})\}_{k=1}^{K_D}. \tag{11}$$

For depth d, width w, and description budget ℓ, define the certified support

$$\mathcal{M}_t(d, w, \ell) = \{\mu \in \text{cl}_{\circ,\text{if},\text{rec},\text{test},\text{tool},\text{sem},\text{src}}(\mathcal{D}_t \cup \mathcal{P}_t^{prim}) : L_{\mathcal{D}_t}(\mu) \le \ell,\ d(\mu) \le d,\ w(\mu) \le w, \nu(\mu) \ne \varnothing\}. \tag{12}$$

The architecture is intelligent only over this closure. Its supported certified utility and support gap are

$$I_t(\mathcal{P}) = \mathbb{E}_{T \sim \mathcal{P}}\left[\max_{\mu \in \mathcal{M}_t(d,w,\ell)} U_\Gamma(T, \mu) \mathbf{1}\{\text{Cert}(\mu, T) \ge V_\Gamma\}\right], \tag{13}$$

$$G_t(\mathcal{P}) = \mathbb{E}_{T \sim \mathcal{P}}\left[\max_{\mu \in \mathcal{M}^*} U_\Gamma(T, \mu) - \max_{\mu \in \mathcal{M}_t(d,w,\ell)} U_\Gamma(T, \mu)\right]_+, \tag{14}$$

$$G_t^{\text{NL}}(\mathcal{P}) = \mathbb{E}\left[\Delta_{ctx}(T) + \Delta_{sem}(T) + \Delta_{src}(T) + \Delta_{rebut}(T) + \Delta_{real}(T) + \Delta_{mode}(T) + \Delta_{ent}(T)\right]_+. \tag{15}$$

Thus more parameters do not by themselves increase intelligence; support expands only by adding primitives, typed compositions, context formalisms, semantic atoms, entailment tests, source/rebuttal-recall mechanisms, realization checkers, verifiers, memories, or tool contracts that reduce $G_t + G_t^{\text{NL}}$ under audit.
The active architecture state contains a bottleneck vector rather than a scalar capability score:

$$\text{Bott}_t(T) = \left(G_t, G_t^{\text{NL}}, \epsilon_{support}, \epsilon_{sub}, \epsilon_{ctx}, \epsilon_{sem}, \epsilon_{src}, \epsilon_{rebut}, \epsilon_{real}, \epsilon_{dep}, d_{shift}, b_{mech}, H_\Gamma, \mathbf{r}_{cum}, \Delta\,\text{Front}^-\right). \tag{16}$$

A repair operator $\rho \in \mathcal{R}_{arch}$ is an admissible primitive, verifier, parser, retriever, memory capsule, causal test, macro, renderer checker, safety rule, or training-data acquisition. It is applied only when its held-out marginal certified value is positive under all current dual prices:

$$\rho_t^* = \arg\max_{\rho \in \mathcal{R}_{arch}} \frac{\text{LCB}_{1-\delta_\rho}\{\Delta S_{\text{ver}}(\rho) - \lambda_B \|\Delta\,\text{Bott}_t^+(\rho)\|_1 - \boldsymbol{\lambda}_R^\top \Delta\mathbf{r}(\rho) - \lambda_M \Delta C_{maint}\}}{c_{impl}(\rho) + c_{train}(\rho) + c_{cal}(\rho) + c_{exact}(\rho)}, \tag{17}$$

$$\text{Apply}(\rho_t^*) = \mathbf{1}\{\text{LCB}_{1-\delta_\rho}(\Delta S_{\text{ver}}) > 0 \wedge \Delta\,\text{Bott}_t^+ \preceq 0 \wedge \Delta\mathbf{r} \preceq \mathbf{b}_R - \mathbf{r}_{cum}\}. \tag{18}$$

This rule changes the architecture at the weakest certified bottleneck: a compression failure triggers factorization or residual coding, a support failure triggers exact rehydration or a new primitive, a language failure triggers context/semantic/source/rebuttal/realization machinery, an intelligence failure triggers new tests or tools, a safety failure tightens authority/risk structure, and a training inefficiency triggers compute reallocation. Parameter scaling is one possible repair only when it has positive lower-confidence frontier movement after these charges.

Novelty is an expansion operator with witnessed boundary failures:

$$\begin{aligned}\mathsf{Expand}(T) = 1 \iff & \Pr_{q_t}[S_\Gamma = 1 \mid \mathcal{M}_t] < \epsilon_s, \quad \mathrm{LCB}_{1-\delta}(\Delta S^{new}_{\mathrm{ver}} - \lambda_c \Delta\mathrm{Cost} - \lambda_R \Delta\mathrm{Risk}) > 0, \\ & \exists \mathcal{W}_T : \forall \mu \in \mathcal{M}_t(d, w, \ell)\ \exists x \in \mathcal{W}_T\ \mathrm{Cert}(S_\Gamma(\mu, x) = 0) \geq \theta_{bdry}. \end{aligned} \tag{19}$$

Without boundary witnesses the system widens search or asks; it does not infer that the world is impossible merely because the current DSL failed.

An adaptive claim family at round r is

$$\mathcal{J}_r = \{h_{rj} : h_{rj} \text{ is } \mathcal{F}_{r-1}\text{-measurable}\}, \qquad \sum_{r,j} \alpha_{rj} \leq \alpha_\Gamma. \tag{20}$$

The requirement that h_{rj} is measurable before the certifying sample is drawn is the anti-p-hacking condition; a hypothesis generated from a test split cannot be certified by that split.

A verifier is

$$v = (\mathcal{X}_v, c_v, n_v, f_v, \rho_v, \psi_v, \kappa_v, \mathsf{train}_v, \mathsf{cal}_v, \mathsf{audit}_v, \mathsf{gen}_v), \tag{21}$$

with domain, cost, calibration trials, observed false passes, dependence embedding, scope penalty, staleness/adaptivity penalty, training split, calibration split, audit split, and generator identity. For a false claim h and an adaptively chosen test x_r, let $X_{v,r}(h, x_r) \in \{0, 1\}$ be pass/fail. The verifier is usable only on an event $\mathcal{C}_{v,r}$ with $\Pr(\mathcal{C}_{v,r}) \geq 1 - \delta_{v,r}$ such that

$$\Pr_0(X_{v,r} = 1 \mid \mathcal{F}_{r-1}, h \text{ false}, \mathcal{C}_{v,r}) \leq p^+_{v,r}(T), \tag{22}$$

where

$$p^+_{v,r}(T) = \min\{1, \mathrm{UCB}^{\mathrm{seq}}_{1-\delta_{v,r}}(p_v; n_v, f_v) + \psi_v(T) + \kappa_v + D_{drift}(v, T) + D_{select}(v, T)\}, \qquad \sum_{v,r} \delta_{v,r} \leq \delta_{cal}. \tag{23}$$

The confidence sequence rather than a fixed-binomial interval is required because verifier choice, replay frequency, and stopping time are adaptive [13]. Conditional residual calibration is required for correlated tests; duplicate passes do not accumulate evidence.

Effective verifier weight is

$$\eta_{v,r} = \eta_{scope}\eta_{stale}\eta_{split}\eta_{dep}\eta_{gen}\eta_{audit} \in [0, 1], \tag{24}$$

where $\eta_{split} = 0$ if any certification event violates Π_{split}, $\eta_{audit} = 0$ if candidate selection inspected the audit trace, $\eta_{gen} = 0$ if the verifier family generated the candidate and has no independent replay split, and $\eta_{dep} = 0$ for duplicate or correlated verifiers without conditional residual evidence. Let $\mathcal{G}_r$ be the dependence partition induced by shared generator, data split, retrieval index, training run, calibration source, prompt, tool, and failure mode. For a block $g \in \mathcal{G}_r$, either a residual pass variable $X^\perp_{g,r}$ is calibrated conditionally on all earlier block outputs or the block contributes only its strongest single calibrated pass:

$$E_{g,r}(h) = 1 + \eta_{g,r}\left(\frac{X^\perp_{g,r}(h)}{p^+_{g,r}(T)} - 1\right), \qquad \eta_{g,r} = 0 \quad \text{if } X^\perp_{g,r} \text{ is unavailable}, \tag{25}$$

$$E_{\mathrm{Dep},r}(h) = \prod_{g \in \mathcal{G}_r} E_{g,r}(h), \qquad \epsilon_{dep} = \mathrm{UCB}^{\mathrm{seq}}_{1-\delta_{dep}} \Pr\{\exists g : \eta_{g,r} = 0 \text{ but treated as independent}\} + D^{dep}_{select} + D^{dep}_{shift}. \tag{26}$$

The claim martingale is formed from dependence blocks, not raw verifier calls:

$$E_{v,r}(h) = 1 + \eta_{v,r}\left(\frac{X_{v,r}(h, x_r)}{p^+_{v,r}(T)} - 1\right), \qquad M_r(h) = \prod_{s \leq r : h \in \mathcal{J}_s} E_{\mathrm{Dep},s}(h). \tag{27}$$

If a verifier fails, $E_{v,r} \leq 1$; if independence, split hygiene, scope, or residual calibration is absent, $\eta_{v,r} = 0$ and the pass contributes exactly one. The certificate after family spending is

$$\mathrm{Cert}(h, T, V_{1:r}) = [\log M_r(h) - \log(1/\alpha_h)]_+, \qquad \alpha_h > 0, \quad \sum_h \alpha_h \leq \alpha_\Gamma. \tag{28}$$

Theorem 1 (Anytime, adaptive, family-valid false-pass control)**.** *On $\mathcal{C} = \cap_{v,r}\mathcal{C}_{v,r}$, if every invoked verifier satisfies $\mathbb{E}_0[E_{v_r,r}(h) \mid \mathcal{F}_{r-1}] \leq 1$ for every false $h \in \mathcal{J}_r$, then $M_r(h)$ is a nonnegative test supermartingale [12]. For every $u \geq 0$,*

$$\Pr_0(\exists h \in \cup_r \mathcal{J}_r : \mathrm{Cert}(h, T, V_{1:\infty}) \geq u) \leq \delta_{cal} + e^{-u}\sum_h \alpha_h. \tag{29}$$

Thus a rendered family with $\sum_h \alpha_h \leq \alpha_\Gamma$ has anytime false-pass probability at most $\delta_{cal} + e^{-u}\alpha_\Gamma$ at excess level u.

Proof. Conditional validity gives the supermartingale property. Ville's inequality yields $\Pr_0(\sup_r M_r(h) \geq e^u/\alpha_h \mid \mathcal{C}) \leq \alpha_h e^{-u}$. Union over the ledgered family and add $\Pr(\mathcal{C}^c) \leq \delta_{cal}$. □

A certificate is scoped:

$$\begin{aligned}\mathrm{Cert}(h,T) = \min\{&\mathrm{Cert}_{obs}, \mathrm{Cert}_{cf}, \mathrm{Cert}_{auth}, \mathrm{Cert}_{priv}, \mathrm{Cert}_{risk}, \mathrm{Cert}_{scope}, \mathrm{Cert}_{support}, \mathrm{Cert}_{dep},\\ &\mathrm{Cert}_{sub}, \mathrm{Cert}_{sem}, \mathrm{Cert}_{ctx}, \mathrm{Cert}_{src}, \mathrm{Cert}_{rebut}, \mathrm{Cert}_{real}, \mathrm{Cert}_{dom}\},\end{aligned} \tag{30}$$

$$\mathrm{Cert}_{dom} = \begin{cases}\infty, & \mathrm{DomOK}(T,\Gamma) = 1,\\ 0, & \text{otherwise},\end{cases} \tag{31}$$

$$\mathrm{Cert}_{dep} = [-\log(\epsilon_{dep} + \epsilon_0)]_+, \tag{32}$$

$$\mathrm{Cert}_{risk} = \min_{j \in \mathcal{J}_R} [-\log((r_{cum,j}/(B_{R,j} + \epsilon)) + \epsilon_0)]_+. \tag{33}$$

Non-applicable caps are set to ∞ by the contract. Evidence level E_k is $\theta_k \le \mathrm{Cert} < \theta_{k+1}$. Every rendered claim, action, memory write, and declassification has a required θ and a family-spending charge in the present scope. Raw pass counts, neural confidence, majority vote, and repeated correlated tests are never evidence.

3 Verifier-native neural substrate

The neural base model is VNS_θ, a proposal and prediction substrate whose public interface is typed state, not hidden text logits. Observations from text, pixels, grids, proof states, tools, tables, code, and environment traces are encoded as a typed event hypergraph

$$x_t = \mathrm{Enc}_\theta(o_t, \Gamma) = \big(\{e_{tm}, \tau_{tm}, \lambda_{tm}, P_{tm}\}_{m \le M_t}, R_t^{obs}\big), \tag{34}$$

where τ is type, λ authority/data taint, P provenance, and R^{obs} observed relations. The substrate state is

$$\begin{aligned}H_t^\theta &= \Big(h_t^{ssm}, \{s_{ti}\}_{i \le N_t}, R_t^{kv}, \mathcal{A}_t^{act}, \mathcal{A}_t^{arc}, \mathcal{G}_t^\theta, D_t^\theta, E_t^{sem,\theta}, C_t^{ctx,\theta}, R_t^{src,\theta}, R_t^{rebut,\theta}, \Omega_t^{real,\theta}\Big),\\ s_{ti} &= (z_{ti}, \pi_{ti}, \tau_{ti}, a_{ti}, p_{ti}, q_{ti}, \kappa_{ti}, \rho_{ti}^{sem}, \rho_{ti}^{ctx}, \rho_{ti}^{src}, \rho_{ti}^{real}).\end{aligned} \tag{35}$$

with long-context recurrent state h^{ssm}, object/event slots s_i, episodic key-value memory R^{kv}, active/archival mechanisms, a differentiable shadow of the e-graph, discourse state, semantic atoms, context commitments, source-coverage state, rebuttal-coverage state, and realization obligations. The update is selective recurrent plus sparse typed binding:

$$h_t^{ssm} = \mathrm{SSM}_\theta\big(h_{t-1}^{ssm}, \sum_m W_{\tau_{tm}} e_{tm}, u_{t-1}, \Gamma\big), \tag{36}$$

$$\tilde{s}_{ti} = s_{t-1,i} + F_\theta\left(s_{t-1,i}, h_t^{ssm}, \sum_{m \in \mathcal{B}_i} A_{im} e_{tm}, \Gamma\right), \tag{37}$$

$$\bar{m}_{ti} = \sum_{(j,k) \in \mathrm{TopK}} \mathrm{Attn}_\theta(\tilde{s}_{ti}, \tilde{s}_{tj}, d_k, h_t^{ssm}, \Gamma), \qquad s_{ti} = \Pi_{type,auth}(\tilde{s}_{ti} + G_\theta(\bar{m}_{ti})). \tag{38}$$

Birth/death of slots is a posterior operation with lost-mass accounting. Authority and provenance labels are joined monotonically during neural updates, so the substrate cannot manufacture instruction or capability labels.
Projection from substrate to mechanisms is explicit:

$$\begin{aligned}\mathrm{Proj}_\theta(H_t^\theta, T, \Gamma) = (&q_\theta(c, \mathfrak{n}, \hat{\Gamma}, A, P \mid T),\ q_\theta([\mu] \mid H_t^\theta, \Gamma),\ \hat{\epsilon}_{support},\ \hat{\epsilon}_{sub},\ \hat{\epsilon}_{sem},\ \hat{\epsilon}_{src},\ \hat{\epsilon}_{ctx},\\ &\hat{\epsilon}_{rebut},\ \hat{\epsilon}_{real},\ \hat{p}_v^+,\ \hat{\eta}_v,\ \widehat{\mathrm{EVC}},\ \hat{U}_\Gamma,\ \hat{H}_\Gamma,\ \hat{\rho}_{reuse},\ \hat{b}_{mech},\ \hat{r}_{cum}).\end{aligned} \tag{39}$$

These quantities choose proposals, routing, widening, and questions. They are never certificates. Exact execution, independent verifiers, calibrated support recall, and gates determine certification.
The base model competes with transformers and state-space models by containing them as proposal subcases while adding typed binding and executable projection. In compact notation,

$$\mathrm{VNS}_\theta = \mathrm{ExecProj} \circ \mathrm{Bind}_{type,auth} \circ \mathrm{Attn}_{sparse} \circ \mathrm{SSM}_{sel} \circ \mathrm{Enc}_\theta\,. \tag{40}$$

With all tokens as slots, full attention, identity executable projection, and disabled gates, this contains a standard transformer proposal class [1]; with a single stream, no slot births, and no attention it contains selective state-space proposal classes [4]. Slot binding supplies variable identity and perceptual object permanence [9]; predictive-state losses supply world-model semantics rather than next-token fluency alone [10, 11].
World modeling is over tests and interventions, not only sequences. For a test $q = (a_{1:k}, \varphi)$,

$$\hat{P}_\theta(q \mid H_t^\theta) = P_\theta\{\varphi(O_{t+1:t+k}) = 1 \mid do(a_{1:k}), H_t^\theta, \Gamma\}, \tag{41}$$

and the substrate must predict observations, counterfactuals, verifier outcomes, conversation normal forms, semantic parses, context/reference commitments, entailment obligations, source coverage, material rebuttals, round-trip realization, support

recall, and executable traces:

$$L_{base} = L_{obs} + \lambda_{psr} L_{psr} + \lambda_{cf} L_{cf} + \lambda_{bind} L_{bind} + \lambda_{memref} L_{memref} + \lambda_{proj} L_{proj} + \lambda_{route} L_{dom} + \lambda_{ctx} L_{ctx}^{base} + \lambda_{sem} L_{sem}^{base} + \lambda_{ent} L_{ent}^{base} + \lambda_{src} L_{src}^{base} + \lambda_{rebut} L_{rebut}^{base} + \lambda_{real} L_{real}^{base}, \tag{42}$$

$$L_{psr} = -\sum_{q \in \mathcal{Q}_t^{audit}} \log \hat{P}_\theta(q \mid H_t^\theta), \qquad L_{proj} = -\log \sum_{\mu \in \mathcal{M}^*(T)} q_\theta([\mu] \mid H_t^\theta, \Gamma), \tag{43}$$

$$L_{memref} = -\log \Pr_\theta\{\text{retrieves every audited certifying parent of } \tau \mid H_t^\theta\}, \tag{44}$$

$$L_{dom} = \mathrm{CE}(\widehat{\mathrm{Route}}_\theta, \mathrm{Route}) + \lambda_{cost}\|\hat{c}_v - c_v\|_1 + \lambda_{supp}\|\hat{\epsilon}_{support} - \epsilon_{support}\|_1 + \lambda_{sub}\|\hat{\epsilon}_{sub} - \epsilon_{sub}\|_1 + \lambda_{sem}\|\hat{\epsilon}_{sem} - \epsilon_{sem}\|_1 + \lambda_{src}\|\hat{\epsilon}_{src} - \epsilon_{src}\|_1 + \lambda_{ctx}\|\hat{\epsilon}_{ctx} - \epsilon_{ctx}\|_1 + \lambda_{rebut}\|\hat{\epsilon}_{rebut} - \epsilon_{rebut}\|_1 + \lambda_{real}\|\hat{\epsilon}_{real} - \epsilon_{real}\|_1 + \lambda_{risk}\|\hat{r}_{cum} - r_{cum}\|_1. \tag{45}$$

Language modeling and perceptual reconstruction are auxiliary views of the same state:

$$L_{obs} = -\sum_j w_j \log p_\theta(o_{tj} \mid H_{<t}^\theta, \Gamma), \quad w_j = w_{claim}\mathbf{1}\{P_j \neq \varnothing\} + w_{style}\mathbf{1}\{j \text{ nonclaim}\} + w_{neg}\mathbf{1}\{j \text{ unsupported claim}\}. \tag{46}$$

The renderer may use dense language capacity for style, compression, and nonclaim glue, but factual, mathematical, executable, private, advice-like, and side-effecting spans require semantic atoms, ledger parents, and entailment witnesses.

Substrate recall is calibrated exactly as other pruning operations. Let $\mathcal{T}_{cert}(T)$ be the set of teacher traces meeting the contract and let $\mathsf{miss}_\theta(T) = 1$ if the substrate plus retrieval assigns no candidate in the bounded exact rehydration neighborhood of any $\tau \in \mathcal{T}_{cert}(T)$. Then

$$\epsilon_{sub}(T) = \mathrm{UCB}^{\mathrm{seq}}_{1-\delta_{sub}}\Big(\Pr\{\mathsf{miss}_\theta(T) = 1\}; \mathcal{D}_{sub}^{audit}\Big) + D_{shift}^{sub}(T) + D_{select}^{sub}(T), \qquad \mathrm{Cert}_{sub} = [-\log(\epsilon_{sub}(T) + \epsilon_0)]_+. \tag{47}$$

If $\epsilon_{sub} > \epsilon_{sub,max}$, inference must widen exact search, rehydrate archive, ask for narrower input, or abstain. This prevents a fluent but misbound base model from becoming an unobserved bottleneck.

The substrate is also audited for shortcut compression. Let $\mathrm{Rehyd}_K(H_t^\theta)$ enumerate the exact symbolic/e-graph/program states within edit, retrieval, and macro-expansion radius K of the neural projection. For a certifying trace family $\mathcal{T}_{cert}(T)$ define

$$\mathrm{Crit}(T) = \{e : \exists \tau \in \mathcal{T}_{cert}(T) \text{ such that removing edge } e \text{ changes } \mathrm{Gate}(\tau) \text{ or } S_\Gamma(\tau)\}, \tag{48}$$

$$\epsilon_{crit}(T) = \mathrm{UCB}^{\mathrm{seq}}_{1-\delta_{crit}} \Pr\{\mathrm{Crit}(T) \not\subseteq \mathrm{Rehyd}_K(H_t^\theta)\} + D_{shift}^{crit} + D_{select}^{crit}. \tag{49}$$

The neural state may be compressed, but critical edges may not be compressed away without an exact reconstruction path. The execution gate uses $\min\{\mathrm{Cert}_{sub}, [-\log(\epsilon_{crit} + \epsilon_0)]_+\}$, and training includes counterfactual edge probes that delete or swap critical parents.

4 Natural-language contractization, rebuttal, and semantic realization

Natural language enters through a posterior over conversation normal forms, not through a privileged sequence model. For dialogue $U_{0:t}$,

$$q_\psi(\mathfrak{n} \mid U_{0:t}, H, \Gamma) = \{(\mathfrak{n}_k, w_k, b_k^{ctx}, b_k^{sem}, b_k^{real})\}_{k \le K_n}, \qquad \sum_k w_k = 1, \tag{50}$$

where

$$\mathfrak{n} = \mathrm{CNF}(U_{0:t}) = (\iota, D, C^{ctx}, \mathcal{A}^{sem}, \Delta, \mathcal{R}^{disc}, \mathcal{B}, \mathcal{Q}^{ask}, \mathcal{R}^{real}). \tag{51}$$

Here ι is intent, D discourse referents, C^{ctx} temporal/deictic/granularity/definition commitments, $\mathcal{A}^{sem}$ semantic atoms, Δ dependencies, $\mathcal{R}^{disc}$ dialogue-state transitions, $\mathcal{B}$ scoped background assumptions, $\mathcal{Q}^{ask}$ admissible clarification questions, and $\mathcal{R}^{real}$ realization constraints. Each atom is

$$a = (m, \varphi, \Omega, \tau, \mathrm{refs}, \mathrm{deps}, V_a, R_a, \alpha_a, \lambda_a, \kappa_a, \gamma_a, \chi_a, \upsilon_a), \tag{52}$$

where $m \in \mathcal{M}_{mode}$ is the speech-act/evidence mode, φ the typed proposition or executable object, Ω its scope, τ its type, κ_a its context key, γ_a its granularity/vagueness class, χ_a its contradiction-search obligation, and υ_a its realization class. The mode set is partitioned as

$$\mathcal{M}_{cert} = \{\text{fact}, \text{math}, \text{code}, \text{private}, \text{action}, \text{quote}, \text{advice}, \text{definition}\}, \tag{53}$$

$$\mathcal{M}_{soft} = \{\text{style}, \text{creative}, \text{preference}, \text{uncertainty}, \text{question}, \text{refusal}, \text{social}, \text{transition}\}. \tag{54}$$

Atoms in $\mathcal{M}_{cert}$ require entailment, source, rebuttal, authority, privacy, context, support, substrate, family-spending, and realization certificates. Atoms in $\mathcal{M}_{soft}$ require mode separation, influence safety, privacy, and no hidden certified-mode atom; they do not become factual claims by fluent wording.

Context loss is separated from semantic loss because many open-ended language failures are not false propositions but unresolved references, shifting time, implicit comparison classes, vague predicates, or missing definitions. Let $C_\star^{ctx}$ be the audited context normal form and $\preceq_{ctx}$ the refinement order induced by Π_{ctx}. Define

$$H_{ctx}(T) = H_q(\iota, D, C^{ctx} \mid U, H, \Gamma), \tag{55}$$

$$\epsilon_{ctx}(T) = \min\{1, \mathrm{UCB}^{\mathrm{seq}}_{1-\delta_{ctx}}(\Pr\{C_\star^{ctx} \not\preceq_{ctx} C_k^{ctx} \text{ for all retained } k\}; \mathcal{D}_{ctx}^{audit}) + b^{ctx} + D_{shift}^{ctx}(T) + D_{select}^{ctx}(T)\}, \tag{56}$$

$$\mathrm{Cert}_{ctx}(T) = [-\log(\epsilon_{ctx}(T) + \epsilon_0)]_+, \tag{57}$$

$$\mathrm{CtxOK}(T, \Gamma) = \mathbf{1}\{H_{ctx} \le h_{ctx} \wedge \epsilon_{ctx} \le \epsilon_{ctx,max} \wedge \mathrm{RefOK}(D) \wedge \mathrm{ScopeOK}(C^{ctx}, \Gamma) \wedge \mathrm{GranOK}(\mathcal{A}^{sem})\}. \tag{58}$$

For a certified-mode atom,

$$\mathrm{Vague}(a) = 1 \Rightarrow \mathrm{DefOK}(a, C^{ctx}, \Gamma) \vee m(a) \in \{\text{question}, \text{uncertainty}, \text{refusal}\}; \tag{59}$$

undefined vague predicates therefore force a definition, scope reduction, question, or uncertainty span. They cannot be silently compiled into a crisp factual answer.

Semantic contractization is admissible only when the posterior has retained the plausible meanings and the retained meanings refine the context commitments:

$$H_{sem}(T) = H_q(\iota, D, \mathcal{A}^{sem}, \Delta \mid U, H, \Gamma), \tag{60}$$

$$\epsilon_{sem}(T) = \min\{1, \mathrm{UCB}^{\mathrm{seq}}_{1-\delta_{sem}}(\Pr\{\mathfrak{n}^\star \notin \mathrm{supp}_{K_n} q_\psi\}; \mathcal{D}^{audit}_{sem}) + b^{sem} + D^{sem}_{shift}(T) + D^{sem}_{select}(T)\}, \tag{61}$$

$$\mathrm{Cert}_{sem}(T) = [-\log(\epsilon_{sem}(T) + \epsilon_0)]_+, \tag{62}$$

$$\mathrm{SemOK}(T, \Gamma) = \mathbf{1}\{H_{sem} \le h_{sem} \wedge \epsilon_{sem} \le \epsilon_{sem,max} \wedge \mathrm{CtxOK}(T, \Gamma) \wedge \mathrm{ModeSep}(\mathcal{A}^{sem}) \wedge \mathrm{DepOK}(\Delta)\}. \tag{63}$$

If SemOK $= 0$ or CtxOK $= 0$, the permitted actions are to ask, narrow, or produce an explicitly scoped uncertainty/refusal span; no factual answer is rendered. The clarification policy maximizes certified entropy reduction rather than conversational fluency:

$$q^* = \arg\max_{q \in \mathcal{Q}_{ask}} \frac{H_{ctx} + H_{sem} - \mathbb{E}_o[H_{ctx} + H_{sem}](U_{0:t} \cup (q, o)) + \lambda_c \mathbb{E}_o \Delta\,\mathrm{DomOK} - \lambda_A \mathrm{Annoy}(q)}{c(q) + \lambda_H H_{bits}(q)}. \tag{64}$$

For a factual or advice-like atom a, retrieval must search both for support and for material rebuttal. Let $\mathcal{S}^\star_{a,+}$ be minimal sufficient support sets and $\mathcal{S}^\star_{a,-}$ material contradiction or scope-defeater sets under R_a, Ω_a, C^{ctx}. For retrieved support $R^+_t(a)$ and rebuttal set $R^-_t(a)$,

$$\mathsf{miss}_{src}(a) = \mathbf{1}\{\forall S \in \mathcal{S}^\star_{a,+} : S \not\subseteq R^+_t(a)\}, \tag{65}$$

$$\mathsf{miss}_{rebut}(a) = \mathbf{1}\{\exists S \in \mathcal{S}^\star_{a,-} : S \not\subseteq R^-_t(a)\}, \tag{66}$$

$$\epsilon_{src}(a) = \min\{1, \mathrm{UCB}^{\mathrm{seq}}_{1-\delta_{src}}(\Pr\{\mathsf{miss}_{src}(a) = 1\}; \mathcal{D}^{audit}_{src}) + D^{src}_{stale}(a) + D^{src}_{select}(a) + D^{src}_{shift}(a)\}, \tag{67}$$

$$\epsilon_{rebut}(a) = \min\{1, \mathrm{UCB}^{\mathrm{seq}}_{1-\delta_{rebut}}(\Pr\{\mathsf{miss}_{rebut}(a) = 1\}; \mathcal{D}^{audit}_{rebut}) + D^{rebut}_{stale}(a) + D^{rebut}_{select}(a) + D^{rebut}_{shift}(a)\}, \tag{68}$$

$$\mathrm{Cert}_{src}(a) = [-\log(\epsilon_{src}(a) + \epsilon_0)]_+, \qquad \mathrm{Cert}_{rebut}(a) = [-\log(\epsilon_{rebut}(a) + \epsilon_0)]_+. \tag{69}$$

For exact internal objects, such as arithmetic traces or proof terms, R_a may set $\epsilon_{src}(a) = \epsilon_{rebut}(a) = 0$ and shift the burden to execution and proof verifiers. For open-domain facts, source recall plus contradiction recall is mandatory: retrieved text proposes parents, and absence of visible contradiction is not evidence unless the contradiction-recall bound is small.

Entailment is explicit. Given ledger parents P_a, an entailment witness e_a, and atom $a = (m, \varphi_a, \ldots)$,

$$\mathrm{Cert}_{ent}(a) = \mathrm{Cert}\big(\mathrm{Ent}(P_a, e_a, \Omega_a, C^{ctx}) \Rightarrow \varphi_a\big), \tag{70}$$

$$\mathrm{RebutOK}(a) = \mathbf{1}\{m(a) \notin \{\text{fact}, \text{advice}, \text{quote}\} \vee \mathrm{Cert}_{rebut}(a) \ge \theta_{rebut}(m)\}, \tag{71}$$

$$\mathrm{Cert}_{atom}(a) = \min\{\mathrm{Cert}_{ctx}, \mathrm{Cert}_{sem}, \mathrm{Cert}_{ent}(a), \mathrm{Cert}_{src}(a), \mathrm{Cert}_{rebut}(a), \min_{p \in P_a} \mathrm{Cert}(p), \mathrm{Cert}_{support}, \mathrm{Cert}_{sub}, \mathrm{Cert}_{dom}, \mathrm{Cert}_{auth}, \mathrm{Cert}_{priv}\}, \tag{72}$$

$$\mathrm{EntOK}(a, Z, \Gamma) = \mathbf{1}\{P_a \ne \varnothing \wedge \mathrm{Cert}_{ent}(a) \ge \theta_{ent}(m) \wedge \mathrm{RebutOK}(a) \wedge \mathrm{Cert}_{atom}(a) \ge V_a\}. \tag{73}$$

Paraphrase, synthesis, advice, and summarization are proof obligations over atoms. A source can be quoted only if span alignment and scope are certified:

$$\mathrm{CiteOK}(a) = \mathbf{1}\{m = \text{quote} \Rightarrow \mathrm{Align}(y_a, P_a) \wedge \mathrm{NoContextLoss}(y_a, P_a, \Omega_a, C^{ctx})\}. \tag{74}$$

Rendering is a semantic realization problem with an independent round-trip checker. For a candidate text y intended to realize atom multiset A_y, let $\mathrm{Atom}(y)$ be the atom multiset extracted from y by a checker not used to generate y. With $\preceq_{sem}$ the semantic refinement order,

$$\epsilon_{real}(y) = \min\{1, \mathrm{UCB}^{\mathrm{seq}}_{1-\delta_{real}}(\Pr\{\neg(\mathrm{Atom}(y) \preceq_{sem} A_y) \vee \mathrm{Hidden}_{cert}(y)\}; \mathcal{D}^{audit}_{real}) + b^{real} + D^{real}_{shift}(y) + D^{real}_{select}(y)\}, \tag{75}$$

$$\mathrm{Cert}_{real}(y) = [-\log(\epsilon_{real}(y) + \epsilon_0)]_+, \tag{76}$$

$$\mathrm{RealOK}(y, A_y, Z, \Gamma) = \mathbf{1}\{\epsilon_{real}(y) \le \epsilon_{real,max} \wedge \mathrm{Atom}(y) \preceq_{sem} A_y \wedge \neg\,\mathrm{Hidden}_{cert}(y) \wedge \Pi_{\mathrm{real}}(y, A_y, \Gamma)\}. \tag{77}$$

Missing evidence is not silently converted into fluent prose:

$$\mathrm{Def}(y) = \{a \in \mathrm{Atom}(y) : m(a) \in \mathcal{M}_{cert},\ \mathrm{EntOK}(a, Z, \Gamma) = 0 \vee \mathrm{CiteOK}(a) = 0 \vee \mathrm{RebutOK}(a) = 0\}, \tag{78}$$

$$\mathrm{ModeOK}(y, Z, \Gamma) = \mathbf{1}\{\forall a \in \mathrm{Atom}(y) : m(a) \in \mathcal{M}_{cert} \vee \mathrm{NoHiddenCertAtom}(a) \wedge \Pi_{\mathrm{style}}(a, \Gamma) \wedge \Pi_{\mathrm{infl}}(a, \Gamma)\}, \tag{79}$$

$$\mathrm{Gate}_{\mathrm{NL}}(y, Z, \Gamma) = \mathbf{1}\{\mathrm{SemOK} \wedge \mathrm{CtxOK} \wedge \mathrm{ModeOK}(y, Z, \Gamma) \wedge \mathrm{RealOK}(y, A_y, Z, \Gamma) \wedge \mathrm{Def}(y) = \varnothing \tag{80}$$

$$\wedge\, \Pi_{\mathrm{priv}}(y, \lambda(y), \Gamma) \wedge \alpha(y) \preceq \alpha(\Gamma) \wedge \mathrm{CumRisk}(y) \le B_R(\Gamma)\}. \tag{81}$$

A best-effort answer with incomplete coverage is represented as $y = y_{cert} \oplus y_{unc} \oplus y_{soft}$ where $\mathrm{Atom}(y_{cert})$ passes the gate, $\mathrm{Atom}(y_{soft}) \cap \mathcal{M}_{cert} = \varnothing$, and every missing certified-mode atom is replaced by an uncertainty atom $u(a)$ with

$$\mathrm{Cert}(u(a)) = \mathrm{Cert}\{\text{no admissible certificate for } a \text{ was found within } B, \Pi_{\mathrm{ctx}}, \Pi_{\mathrm{src}}, \Pi_{\mathrm{rebut}}, \Pi_{\mathrm{real}}, \Pi_{\mathrm{dom}}\}. \tag{82}$$

Thus broad language is not solved by making a larger decoder; it is made admissible only by retaining plausible contexts, searching for support and defeaters, and proving that the surface text realizes only the certified atom plan.

Proposition 2 (No unsupported or hidden factual rendering)**.** *Assume the context normalizer, semantic checker, source retriever, rebuttal retriever, and round-trip realization checker have calibrated false-negative bounds* $\epsilon_{ctx}, \epsilon_{sem}, \epsilon_{src}, \epsilon_{rebut}, \epsilon_{real}$*, and entailment certificates are family-valid. If* $\mathrm{Gate}_{\mathrm{NL}}(y, Z, \Gamma) = 1$*, then every certified-mode atom in* y *has scoped certificate at least its required threshold and no extra certified-mode atom is hidden in* y *except on an event of probability at most* $\delta_{ctx} + \delta_{sem} + \delta_{src} + \delta_{rebut} + \delta_{real} + e^{-u}\alpha_\Gamma$ *at excess level* u*.*

Proof. A certified-mode atom can pass $\mathrm{Gate}_{\mathrm{NL}}$ only through EntOK, RebutOK, RealOK, and empty $\mathrm{Def}(y)$. The atom certificate is a minimum over context, semantic, entailment, source, rebuttal, parent, support, substrate, domain, authority, and privacy caps. Hidden factual text is excluded by the independent round-trip event. Context, semantic, source, rebuttal, and realization misses contribute their calibration failure events; entailment false passes are controlled by the family-valid e-value theorem. Union over rendered atoms using the ledgered family budget gives the stated bound. □

5 Compiler, posterior, and support-preserving inference

The compiler preserves alternatives:

$$q_\phi(c, \mathfrak{n}, \hat{\Gamma}, A, P \mid O, H) = \{(c_k, \mathfrak{n}_k, \hat{\Gamma}_k, A_k, P_k, w_k, \Sigma_k, b_k, b_k^{ctx}, b_k^{sem}, b_k^{real})\}_{k=1}^K, \qquad \sum_k w_k = 1, \tag{83}$$

where b_k is an upper bound on mass discarded by parser pruning, b_k^{ctx} by context/reference pruning, b_k^{sem} by semantic-contract pruning, and b_k^{real} by realization-plan pruning. If Γ is fixed externally, only latent fields are inferred. Canonicalization is posterior-valued, not destructive:

$$q(\bar{c} \mid c) = \sum_{g \in Sym(c)} q(g \mid c)\delta_{\mathrm{canon}(g \cdot c)}, \quad q(g \mid c) \propto \exp[-K(g \cdot c) - \lambda_d D(g \cdot c, c_{ref})]. \tag{84}$$

Canonicalization may merge states only with a reversible witness or a support-loss bound

$$\epsilon_{canon} = \Pr_q[S_\Gamma(c) \neq S_\Gamma(\bar{c})] \leq \epsilon_\Gamma. \tag{85}$$

For candidate μ,

$$\begin{aligned} L_{\mathcal{D}}(\mu \mid \Gamma) = \ell_0(\tilde{\mu}) + \sum_{k \in calls(\tilde{\mu})} [-\log p(k \mid \tau, \Gamma)] + \lambda_b B(\mu) + \lambda_\Omega \Omega(\mu) + \lambda_r R_{reg}(\mu) \\ + \lambda_s S_{scope}(\mu) + \lambda_{ctx} S_{ctx}(\mu) + \lambda_{sem} S_{sem}(\mu) + \lambda_{src} S_{src}(\mu) + \lambda_{rebut} S_{rebut}(\mu) + \lambda_{real} S_{real}(\mu) + \lambda_{fp} e^{-\mathrm{Cert}(\mu)}, \end{aligned} \tag{86}$$

$$\begin{aligned} \mathcal{E}(\mu, c, \mathfrak{n}, V; T, \Gamma) = D_\Gamma(Y, \hat{Y}) + \lambda_o D_{obs} + \lambda_{cf} D_{cf} + \beta L_{\mathcal{D}} + \lambda_\chi V_\chi + \lambda_a U_{auth} \\ + \lambda_c \mathrm{Cost} + \lambda_{cal} \mathrm{Cal} + \lambda_{pr} \epsilon_{prune} + \lambda_R \mathrm{Risk} + \lambda_{ctx} \epsilon_{ctx} + \lambda_{sem} \epsilon_{sem} + \lambda_{src} \epsilon_{src} \\ + \lambda_{rebut} \epsilon_{rebut} + \lambda_{real} \epsilon_{real} + \lambda_{ent} V_{ent}. \end{aligned} \tag{87}$$

A trainable score s_θ is a bounded proposal prior, $|s_\theta| \leq s_{max}$:

$$q([\mu], c, \mathfrak{n}, V \mid T, \Gamma) \propto q_\phi(c, \mathfrak{n}, \Gamma \mid T) \exp[-\mathcal{E}([\mu], c, \mathfrak{n}, V; T, \Gamma) + s_\theta([\mu], c, \mathfrak{n}, T)]. \tag{88}$$

It can reduce search cost but cannot reduce $p_{v,r}^+$, increase $\eta_{v,r}$, lower a gate threshold, or remove a lost-support, context-loss, semantic-loss, source-recall, rebuttal-recall, realization-loss, privacy, or cumulative-risk penalty.
At recurrent step r,

$$B^{(r)} = (Z^{(r)}, E^{(r)}, M^{(r)}, Q^{(r)}, V^{(r)}, C^{(r)}, A^{(r)}, G^{(r)}, K^{(r)}, D^{(r)}, E_{sem}^{(r)}, C_{ctx}^{(r)}, R_{src}^{(r)}, R_{rebut}^{(r)}, \Omega_{real}^{(r)}, h^{(r)}). \tag{89}$$

Slot i has type posterior $\pi_i \in \Delta^{|\mathcal{T}|}$ and authority label $a_i \in \mathcal{L}_{auth}$. Sparse interaction is typed, tainted, and budgeted:

$$s_{ijk}^{(r)} = u_\theta(z_i, z_j, m_k, h, \Gamma) + \log m_{type}(i, j, k) + \log m_{auth}(i, j, k) - \lambda_c c_{ijk}, \tag{90}$$

$$\mathcal{N}_i^{(r)} = \mathrm{TopK}_{j,k} s_{ijk}^{(r)}, \qquad m_i^{(r)} = \sum_{(j,k) \in \mathcal{N}_i^{(r)}} \Pr(j, k \mid i, B^{(r)}). \tag{91}$$

Let $R_i^{(r)}$ be the event that every omitted interaction whose inclusion can change S_Γ, context commitments, semantic atoms, source coverage, material rebuttals, realization obligations, or a gate by more than Δ_Γ is absent. The support guard is not the model probability alone but a calibrated recall bound against exact expansions:

$$\hat{\epsilon}_{mass}^{(r)} = \left[\sum_i (1 - m_i^{(r)})\right]_0^1, \tag{92}$$

$$\hat{\epsilon}_{rec}^{(r)} = \mathrm{UCB}_{1-\delta_{rec}}^{\mathrm{seq}}\left(\Pr\{\exists i : R_i^{(r)} = 0\}; \mathcal{D}_{rec}^{exact}\right), \tag{93}$$

$$\epsilon_{prune}^{(r)} = \min\{1, \hat{\epsilon}_{mass}^{(r)} + \hat{\epsilon}_{rec}^{(r)} + D_{shift}^{rec}(T) + D_{select}^{rec}(T)\}, \tag{94}$$

$$\mathrm{Cert}_{support}^{(r)} = \left[-\log(\epsilon_{prune}^{(r)} + \epsilon_0)\right]_+. \tag{95}$$

If $\epsilon^{(r)}_{prune} > \epsilon_{max}(\Gamma)$, the cell must enlarge K, widen the beam, rehydrate archival candidates, recover dropped context/semantic/source/rebuttal/realization alternatives, or enter an exact e-graph/program stage. If it does not, the final certificate is capped by $\mathrm{Cert}^{(r)}_{support}$; for $\epsilon^{(r)}_{prune} = 1$ the cap is zero.

The differentiable update proposes; typed projection, exact execution, and verification dispose:

$$\bar{B}^{(r+1)} = \mathrm{TypedMessage}_\theta(B^{(r)}, C, \Gamma), \qquad B^{(r+1)} = \Pi_{type,auth,budget}\{B^{(r)} + \eta_m(\bar{B}^{(r+1)} - B^{(r)}) - \eta_e \nabla_B \tilde{\mathcal{E}}_\theta(B^{(r)}; T)\}. \tag{96}$$

Resolution is staged,

$$\iota \to \sigma \to \pi \to \eta, \tag{97}$$

where ι are invariants, σ sketches/e-classes, π executable programs/policies, and η optional residuals. Stage $j+1$ is entered iff

$$\begin{aligned}\mathbb{E}[\Delta\mathrm{Cert} + \Delta U_\Gamma \mid j] - \lambda_c \Delta\mathrm{Cost} - \lambda_L \Delta L - \lambda_R \Delta\mathrm{Risk} - \lambda_{supp}\Delta\epsilon_{prune} \\ - \lambda_{ctx}\Delta\epsilon_{ctx} - \lambda_{sem}\Delta\epsilon_{sem} - \lambda_{src}\Delta\epsilon_{src} - \lambda_{rebut}\Delta\epsilon_{rebut} - \lambda_{real}\Delta\epsilon_{real} > 0.\end{aligned} \tag{98}$$

Residuals are escrowed: they may rank proposals, choose repairs, or render uncertainty, but cannot certify claims or authorize side effects unless wrapped by independent verifiers:

$$\mathrm{Cert}(\eta, T) \le \min_{v \in \nu(\eta)} \mathrm{Cert}(v, T), \qquad \mathrm{Gate}(a_\eta, Z, \Gamma) = 0 \text{ if } \alpha(a_\eta) \not\preceq \alpha(\Gamma). \tag{99}$$

Tests are actions over uncertainty. With $\Phi = e^{-\mathrm{Cert}}$,

$$\begin{aligned}e^* = \arg\max_{e \in A_{allow}} \frac{1}{c(e)} \Big\{ & \mathbb{E}_{o \sim p(o|e,T)}[\Delta S_{\mathrm{ver}}(o)] + \lambda_I \mathrm{I}(o; M \mid e, T) + \lambda_V \mathbb{E}[\Phi_r - \Phi_{r+1}] \\ & - \lambda_R \mathrm{Risk}(e) - \lambda_H H_{bits}(e) - \lambda_A \mathrm{Annoy}(e) - \lambda_P \epsilon_{support}(e) \\ & - \lambda_{ctx}\epsilon_{ctx}(e) - \lambda_{sem}\epsilon_{sem}(e) - \lambda_{src}\epsilon_{src}(e) - \lambda_{rebut}\epsilon_{rebut}(e) - \lambda_{real}\epsilon_{real}(e) \Big\}.\end{aligned} \tag{100}$$

The loop halts when the contract threshold is met, a permitted clarification dominates, or

$$\begin{aligned}\mathrm{EVC}_r = \mathbb{E}[\max_b U_\Gamma(b, r+1) - \max_b U_\Gamma(b, r)] - \lambda_F \Delta F - \lambda_L \Delta L - \lambda_R \Delta\mathrm{Risk} - \lambda_P \Delta\epsilon_{prune} \\ - \lambda_{ctx}\Delta\epsilon_{ctx} - \lambda_{sem}\Delta\epsilon_{sem} - \lambda_{src}\Delta\epsilon_{src} - \lambda_{rebut}\Delta\epsilon_{rebut} - \lambda_{real}\Delta\epsilon_{real} \le 0.\end{aligned} \tag{101}$$

6 Verification, falsification, authority, and rendering

Every provisional pass is attacked. For verifier v,

$$x^*_v = \arg\max_{x \in \mathcal{X}_T} \Pr[v(\mu, x) = \mathrm{pass}] - \Pr[S_\Gamma(\mu, x) = 1] - \lambda_x \mathrm{Cost}(x), \tag{102}$$

using enumeration in exact domains and learned adversaries in soft domains. A verifier cannot certify a mechanism family it generated unless an independent replay set or independent verifier family agrees. Calibration records update (n_v, f_v) on near-miss failures; certificates are recomputed after each replay window.

Authority is a lattice $(\mathcal{L}_{auth}, \preceq, \vee)$ with labels for data, retrieved sources, tool outputs, user instructions, developer/system policy, private data, and granted capabilities. All transitions are label monotone. Declassification is proof-carrying:

$$\mathrm{Dec}_\ell(x, Z, \Gamma) = 1 \Rightarrow \mathrm{Cert}(\text{privacy proof}, Z) \ge \theta_{priv}(\ell) \wedge \Pi_{\mathrm{priv}}(x, \ell, \Gamma). \tag{103}$$

External text and tool output are data unless lifted by a valid contract rule:

$$\mathrm{Auth}(\text{tool output}) = \text{data}, \qquad \text{data} \not\succeq \text{instruction}, \qquad \text{data} \not\succeq \text{capability}. \tag{104}$$

Define

$$\mathrm{AuthOK}(a, Z) = \mathrm{Auth}(a, Z) \succeq \alpha(a), \qquad \mathrm{CertOK}(a, Z) = \mathrm{Cert}(a, Z) \ge V_a, \tag{105}$$

$$\begin{aligned}\mathrm{GrantOK}(a, Z, \Gamma) = \mathrm{Rollback}(a) \vee \big(&\mathrm{ExplicitGrant}(a, \Gamma) \wedge \mathrm{Auth}(\text{grant}, Z) \succeq \text{instruction} \\ &\wedge \mathrm{Cert}(\text{grant}, Z) \ge \theta_{grant} \wedge \mathrm{Fresh}(\text{grant}) \wedge \mathrm{ScopeOK}(\text{grant}, a, \Gamma)\big),\end{aligned} \tag{106}$$

$$\mathrm{PrivOK}(a, Z, \Gamma) = \Pi_{\mathrm{priv}}(\mathrm{read}(a) \cup \mathrm{write}(a), \lambda(a), \Gamma) \wedge \mathrm{Exfil}(a, Z) \le E_a, \tag{107}$$

$$\mathrm{SandOK}(a, Z, \Gamma) = \mathrm{Sandbox}(a, \Gamma) \wedge \mathrm{RollbackPlan}(a, Z) \wedge \mathrm{Monitor}(a, Z), \tag{108}$$

$$\mathrm{RiskOK}(a, Z, \Gamma) = \mathbf{1}\{\mathbf{U}_R(a, Z) \preceq \mathbf{B}_R(\Gamma) \wedge \sup_{w \in \mathcal{W}_R} w^\top \mathbf{U}_R(a, Z) \le \sup_{w \in \mathcal{W}_R} w^\top \mathbf{B}_R(\Gamma)\}, \tag{109}$$

$$\mathbf{U}_R(a, Z) = \mathrm{UCB}_{1-\delta_R}(\mathbf{r}_t + \Delta\mathbf{r}(a)), \qquad \Delta\mathbf{r}(a) = (\Delta r^{impact}, \Delta r^{priv}, \Delta r^{exfil}, \Delta r^{cap}, \Delta r^{infl}, \Delta r^{conf}, \Delta r^{model}, \Delta r^{dep}). \tag{110}$$

Cumulative safety is budgeted over the transcript and over channels, not checked only per action and not averaged into a scalar:

$$\mathbf{r}_{t+1} = \mathbf{r}_t + \widehat{\Delta\mathbf{r}}(a_t) - \mathbf{c}^{roll}_t, \tag{111}$$

$$0 \preceq \mathbf{c}^{roll}_t \preceq \mathrm{Cred}(a_t), \qquad \mathrm{Cred}_j(a_t) > 0 \Rightarrow \mathrm{Cert}(\mathrm{monitor}_j \wedge \mathrm{restored}_j, Z_{t+1}) \ge \theta_{roll,j}. \tag{112}$$

A rollback credit cannot restore privacy disclosure, capability transfer, or external influence unless the corresponding external state has a certified reversal witness; otherwise that coordinate is absorbing. The action gate is

$$\begin{aligned}\mathrm{Gate}(a, Z, \Gamma) = \mathbf{1}\{&\mathrm{DomOK}(Z, \Gamma) \wedge \mathrm{IntentOK}(a, Z, \Gamma) \wedge \Pi_{\mathrm{perm}}(a, \Gamma) \wedge \mathrm{AuthOK}(a, Z) \wedge \mathrm{Pre}(a, Z) \wedge \mathrm{CertOK}(a, Z) \\ &\wedge \mathbb{E}[\mathrm{Impact}(a)] \le I_a \wedge \mathrm{Budget}(a) \le B_a \wedge \mathrm{GrantOK}(a, Z, \Gamma) \\ &\wedge \mathrm{PrivOK}(a, Z, \Gamma) \wedge \mathrm{SandOK}(a, Z, \Gamma) \wedge \mathrm{RiskOK}(a, Z, \Gamma) \wedge \Pi_{\mathrm{infl}}(a, \Gamma) \wedge \mathrm{NoConflict}(a, Z, \Gamma)\}.\end{aligned} \tag{113}$$

Explicit grant can satisfy reversibility for scoped irreversible actions only when the grant itself has instruction authority, freshness, scope, and certificate. It cannot bypass permission, authority, preconditions, certificates, impact limits, budget, privacy, sandboxing, cumulative-risk limits, exfiltration bounds, or policy conflict.

Proposition 3 (Data noninterference)**.** *If updates are label-monotone, data labels cannot dominate instruction or capability labels, and* GrantOK *requires instruction-authority grant provenance, then data-only inputs cannot make* $\mathrm{Gate}(a, Z, \Gamma) = 1$ *for an action requiring instruction or capability authority unless the authority is already present in* Γ *or in a pre-existing explicit granted capability.*

Proof. Monotone joins preserve the partial order. No sequence of joins with labels below or incomparable to $\alpha(a)$ can produce a label dominating $\alpha(a)$. Thus AuthOK remains false without an independent grant. Data also cannot manufacture GrantOK because grant provenance must dominate instruction. The gate is conjunctive, so any failed conjunct blocks the action. □

Renderers are semantic transducers from certified state to text. For a span y_j, let $\mathrm{Atom}(y_j)$ be the output atoms and P_a the ledger parents of atom a:

$$\lambda(y_j) = \bigvee_{a \in \mathrm{Atom}(y_j)} \left(\lambda_a \vee \bigvee_{p \in P_a} \lambda(p) \right), \tag{114}$$

$$\mathrm{Cert}(y_j) = \min_{a \in \mathrm{Atom}(y_j) : m(a) \in \mathcal{M}_{cert}} \mathrm{Cert}_{atom}(a), \tag{115}$$

$$\alpha(y_j) = \bigvee_{a \in \mathrm{Atom}(y_j)} \left(\alpha_a \vee \bigvee_{p \in P_a} \alpha(p) \right). \tag{116}$$

A factual sentence, mathematical derivation, private datum, code execution instruction, source quote, advice-like claim, or irreversible plan inherits the maximum evidence, context, source, rebuttal, realization, privacy, and authority requirement of its atoms. A purely stylistic span has no certificate requirement only if the independent checker verifies $\mathrm{Atom}(y_j) \cap \mathcal{M}_{cert} = \varnothing$ and Π_{infl} accepts the span.

$$\begin{aligned} \mathrm{Gate}_{render}(y_j, Z, \Gamma) = \mathbf{1}\{ & \mathrm{DomOK}(Z, \Gamma) \wedge \mathrm{IntentOK}(y_j, Z, \Gamma) \wedge \mathrm{Gate}_{\mathrm{NL}}(y_j, Z, \Gamma) \\ & \wedge \Pi_{\mathrm{priv}}(y_j, \lambda(y_j), \Gamma) \wedge \alpha(y_j) \preceq \alpha(\Gamma) \}. \end{aligned} \tag{117}$$

Unsupported certified-mode spans are replaced by uncertainty, a question, a scoped assumption, or refusal; unsupported soft spans are allowed only when they remain nonclaim text under Π_{style}.

7 Compression, memory, and capacity limits

Memory is two-tier:

$$\mathcal{A}_{act} \subset \mathcal{A}_{arc}, \qquad \zeta = ([\mu], \sigma, \chi, \nu, \mathrm{scope}, \mathrm{failures}, \mathrm{transfer}, L_{\mathcal{D}}, C_{exec}, \mathrm{age}, \mathrm{auth}, \mathrm{drift}, \mathrm{ctx}, \mathrm{sem}, \mathrm{src}, \mathrm{rebut}, \mathrm{real}, \mathrm{mode}). \tag{118}$$

Only $\mathcal{A}_{act}$ participates in low-latency inference; archival items are retrieval evidence, replay material, and sleep-phase candidates. The library objective is

$$\begin{aligned} \min_{\mathcal{D}, \mathcal{A}} \quad & K(\mathcal{D}) + K(\mathcal{A}) + \sum_i L_{\mathcal{D},\mathcal{A}}(\tau_i \mid \Gamma_i) + \lambda_{reg}\mathrm{Regress} + \lambda_{maint}\mathrm{Maint} \\ & + \lambda_{ent} H(\mathcal{A}_{act}) + \lambda_{auth}\mathrm{CapRisk} + \lambda_{ctx}\mathrm{CtxErr} + \lambda_{sem}\mathrm{SemErr} + \lambda_{src}\mathrm{SrcMiss} + \lambda_{rebut}\mathrm{RebutMiss} + \lambda_{real}\mathrm{RealErr}. \end{aligned} \tag{119}$$

The compressed library is hierarchical:

$$\mathcal{Z}_t = (\mathcal{D}_t^{prim}, \mathcal{D}_t^{local}, \mathcal{D}_t^{global}, \mathcal{D}_t^{ctx}, \mathcal{D}_t^{sem}, \mathcal{D}_t^{src}, \mathcal{D}_t^{safe}, \mathcal{R}_t^{resid}), \tag{120}$$

where active macros encode reusable executable structure, context/semantic/source/safety dictionaries encode reusable interfaces and obligations, and $\mathcal{R}^{resid}$ stores incompressible or unstable residuals with explicit distortion labels. A proposed macro is admitted at the lowest level whose code length plus dispatch entropy improves the held-out rate-distortion frontier:

$$\begin{aligned} \Delta\,\mathrm{Front}_t(m) = \mathrm{LCB}_t^{\mathrm{seq}} & \left[\Delta L + \Delta C_{exec} + \Delta S_{ver} - K(m) - H(\mathrm{dispatch}_m \mid \mathit{scope}) \right] \\ & - \lambda_I \Delta I_m - \lambda_R \boldsymbol{\lambda}_R^\top \Delta \mathbf{r}_m - \lambda_{dep} \Delta \epsilon_{dep,m}, \end{aligned} \tag{121}$$

$$\Delta I_m = \sum_{m' \in \mathcal{A}_{act}} \left| \mathrm{Cov}(g(m), g(m')) \right| + \Pr\{m \text{ shadows a stricter verified primitive}\}. \tag{122}$$

The interference charge prevents the active library from filling with near-duplicate shortcuts that reduce local description length while increasing retrieval entropy, dependency, maintenance, or unsafe capability concentration.
For a candidate macro/capsule m and replay draw $T_i \sim \mathcal{R}_t$, define bounded gain $g_i(m) \in [-G, G]$:

$$\begin{aligned} g_i(m) = & \Delta L_i + \Delta\mathrm{Solve}_i + \Delta\mathrm{Latency}_i + \Delta\mathrm{Reliability}_i + \Delta\mathrm{SampleEff}_i + \Delta\mathrm{Transfer}_i \\ & - K(m) - \eta\Delta\mathrm{Branch}_i - \xi\Delta\mathrm{Ambig}_i - \rho\Delta\mathrm{Risk}_i - \kappa\Delta\mathrm{CalErr}_i \\ & - \nu\Delta\mathrm{Regress}_i - \mu\Delta\mathrm{Maint}_i - \lambda_{fp}\Delta\Phi_i - \lambda_{auth}\Delta\mathrm{CapRisk}_i \\ & - \lambda_{ctx}\Delta\mathrm{CtxErr}_i - \lambda_{sem}\Delta\mathrm{SemErr}_i - \lambda_{src}\Delta\mathrm{SrcMiss}_i - \lambda_{rebut}\Delta\mathrm{RebutMiss}_i - \lambda_{real}\Delta\mathrm{RealErr}_i - \lambda_{mode}\Delta\mathrm{ModeConf}_i. \end{aligned} \tag{123}$$

A macro is not a compressed capability unless it has an expansion witness W_m and a scoped equivalence certificate

$$\mathrm{Cert}_{eq}(m) = \inf_{x \in \Omega_m} \mathrm{Cert}\big(\mathrm{Exec}(m,x) = \mathrm{Exec}(W_m,x) \wedge \Lambda(\mathrm{Exec}(m,x)) \equiv_\Gamma \Lambda(\mathrm{Exec}(W_m,x))\big). \tag{124}$$

Opaque residual shortcuts may rank proposals but are not active macros.

Replay samples and macro candidates are selected adaptively, so admission uses a sequence-valid, multiplicity-corrected radius. With N_t candidates tested up to time t,

$$\widehat{A}_t(m) = \frac{1}{n_t(m)} \sum_{i=1}^{n_t(m)} g_i(m), \qquad \widehat{\sigma}_t^2(m) = \frac{1}{n_t(m)-1} \sum_i (g_i(m) - \widehat{A}_t(m))^2, \tag{125}$$

$$\mathrm{Rad}_t(m) = \sqrt{\frac{2\widehat{\sigma}_t^2(m)\log(3N_t/\delta)}{n_t(m)}} + \frac{3G\log(3N_t/\delta)}{n_t(m)} + D_{adapt}(m), \tag{126}$$

$$\mathrm{LCB}_t^{\mathrm{seq}}\, A(m) = \widehat{A}_t(m) - \mathrm{Rad}_t(m) - D_{drift}(m) - D_{leak}(m) - D_{select}(m), \tag{127}$$

$$\mathrm{UCB}_t^{\mathrm{seq}}\, A(m) = \widehat{A}_t(m) + \mathrm{Rad}_t(m) + D_{drift}(m) + D_{leak}(m) + D_{select}(m). \tag{128}$$

Admission is a decision rule, not a learned preference:

$$\begin{aligned}\mathrm{Admit}_{act}(m) = \mathbf{1}\{\, & \mathrm{LCB}_t^{\mathrm{seq}}\, A(m) > 0 \wedge \mathrm{Cert}(m) \geq \theta_{act} \wedge \mathrm{Cert}_{eq}(m) \geq \theta_{eq} \wedge \mathrm{NoCapEsc}(m) \\ & \wedge \mathrm{ReplayPass}(m) \wedge H(\mathrm{Args}_m \mid scope) \leq h_m \wedge \mathrm{DecodeCost}(m) \leq B_{dec} \\ & \wedge \epsilon_{ctx}(m) \leq \epsilon_{ctx,max} \wedge \epsilon_{sem}(m) \leq \epsilon_{sem,max} \wedge \epsilon_{src}(m) \leq \epsilon_{src,max} \wedge \epsilon_{rebut}(m) \leq \epsilon_{rebut,max} \wedge \epsilon_{real}(m) \leq \epsilon_{real,max} \\ & \wedge \epsilon_{dep}(m) \leq \epsilon_{dep,max} \wedge \Delta\,\mathrm{Front}_t(m) > 0 \wedge \Delta\mathbf{r}(m) \preceq \mathbf{B}_R - \mathbf{r}_t\}.\end{aligned} \tag{129}$$

Demotion is symmetric:

$$\begin{aligned}\mathrm{Demote}(m) = \mathbf{1}\{\, & \mathrm{UCB}_t^{\mathrm{seq}}\, A(m) < 0 \vee \mathrm{Regress}(m) \vee \mathrm{Stale}(m) \vee \mathrm{BranchExplosion}(m) \\ & \vee \mathrm{ScopeMismatch}(m) \vee \mathrm{CapEsc}(m) \vee \mathrm{CtxShift}(m) \vee \mathrm{SemShift}(m) \\ & \vee \mathrm{SrcStale}(m) \vee \mathrm{RebutStale}(m) \vee \mathrm{RealDrift}(m) \vee \mathrm{DepShift}(m) \vee \Delta\,\mathrm{Front}_t(m) < 0\}.\end{aligned} \tag{130}$$

The active set is a constrained selection problem:

$$\mathcal{A}_{act} = \operatorname*{arg\,max}_{S \subset \mathcal{A}_{arc}} \sum_{m \in S} \mathrm{LCB}_t^{\mathrm{seq}}\, A(m) - \lambda_{pair} \sum_{m \neq m'} \mathrm{Overlap}(m,m') - \lambda_{ret} H(R_S) - \lambda_{mode}\mathrm{ModeConf}(S) \tag{131}$$

subject to $C_{lat}(S) \leq B_{lat}$, $C_{mem}(S) \leq B_{mem}$, $\mathrm{CapRisk}(S) \leq B_{cap}$, $\epsilon_{ctx}(S) \leq \epsilon_{ctx,max}$, $\epsilon_{sem}(S) \leq \epsilon_{sem,max}$, $\epsilon_{src}(S) \leq \epsilon_{src,max}$, $\epsilon_{rebut}(S) \leq \epsilon_{rebut,max}$, and $\epsilon_{real}(S) \leq \epsilon_{real,max}$, $\epsilon_{dep}(S) \leq \epsilon_{dep,max}$, $\Delta\mathbf{r}(S) \preceq \mathbf{B}_R - \mathbf{r}_t$, and $\sum_{m \in S} \Delta I_m \leq I_{max}$. Retrieval is approximate but certification is exact; approximate nearest neighbors can miss a useful macro but cannot certify an unsafe or unsupported one.

Proposition 4 (Macro search gain)**.** *If a raw search has branching b and depth L, while macro m changes depth to L' and branching to γb, then search improves only if*

$$b^{L-L'} > \gamma^{L'}. \tag{132}$$

With pruning factors p_d, replace b at depth d by bp_d. Thus syntax compression is insufficient; admission must charge branch, ambiguity, replay, maintenance, and false-pass risk.

Proposition 5 (Sequence-valid macro admission prevents expected degradation)**.** *Assume $g_i(m) \in [-G, G]$, the replay distribution covers the deployment scope up to the drift penalty, and the candidate count is charged by N_t. Uniformly over adaptive macro proposals and stopping times, with probability at least $1-\delta$, every admitted macro has nonnegative amortized gain on its certified scope and is extensionally equivalent to its expansion witness.*

Proof. The empirical-Bernstein confidence sequence gives $A_t(m) \geq \mathrm{LCB}_t^{\mathrm{seq}}\, A(m)$ simultaneously over candidates and stopping times after drift, leakage, and selection charges. Admission requires this lower bound, the equivalence certificate, replay, scope, and capability checks to pass. □

The architecture cannot compress arbitrary additional data. Let $D_{1:n}$ be tasks, $U_{1:n}$ their language interfaces, $A_{1:n}^{sem}$ their semantic atoms, $S_{1:n}^{src}$ their source states, $R_{1:n}^{style}$ their residual style, and $\mathcal{Z}$ the learned library. For any code satisfying contract error ϵ,

$$\mathbb{E}L(D_{1:n} \mid \mathcal{Z}, \Gamma, V) \geq H(D_{1:n} \mid \mathcal{Z}, \Gamma, V) - O(\epsilon n), \tag{133}$$

$$\begin{aligned}\mathbb{E}L(U_{1:n} \mid \mathcal{Z}, \Gamma, V) \geq\ & H(C_{1:n}^{ctx}, A_{1:n}^{sem}, S_{1:n}^{src}, S_{1:n}^{rebut}, R_{1:n}^{real} \mid \mathcal{Z}, \Gamma, V) \\ & + H(R_{1:n}^{style} \mid C_{1:n}^{ctx}, A_{1:n}^{sem}, S_{1:n}^{src}, S_{1:n}^{rebut}, R_{1:n}^{real}, \mathcal{Z}) - O(\epsilon n).\end{aligned} \tag{134}$$

The operational compression frontier is the constrained rate-distortion value

$$\mathcal{R}_t(D,B) = \min_{\mathcal{D},\mathcal{A}} K(\mathcal{D}) + K(\mathcal{A}) + \mathbb{E}[L_{\mathcal{D},\mathcal{A}}(T)] \quad \text{s.t.} \quad \mathbb{E}[d_\Gamma(T,\hat{T})] \leq D,\ C_{lat} \leq B_{lat},\ C_{mem} \leq B_{mem},\ \mathbf{r} \preceq \mathbf{B}_R,\ \epsilon_{dep} \leq \epsilon_{dep,max}. \tag{135}$$

A library expansion is useful only if it moves this frontier downward on a held-out scope; memorizing exceptions moves archival storage, not active compression.

Proposition 6 (Frontier-valid compression)**.** *Assume candidate macros are evaluated on split-clean replay with sequence-valid lower bounds, exact expansion witnesses, and componentwise risk/dependence gates. If every active admission satisfies* $\Delta\,\mathrm{Front}_t(m) > 0$ *and every demoted macro has* $\mathrm{UCB}_t^{\mathrm{seq}}\,\Delta\,\mathrm{Front}_t(m) < 0$ *or a gate violation, then active-memory growth is bounded by the number of positive frontier moves plus the covering number of certified scopes. Additional data that have conditional entropy larger than the best admitted code improvement remain residual and cannot be compressed without increasing distortion or violating a gate.*

Proof. Each admitted macro decreases the held-out rate-distortion LCB after dispatch, interference, maintenance, dependence, and vector-risk charges. A finite memory budget and nonnegative code lengths forbid infinitely many positive decreases above the confidence radius in the same certified scope. If the conditional entropy of new tasks is not reduced by any admissible code, the source-coding lower bound applies to $\mathcal{R}^{resid}$; storing it in $\mathcal{A}_{act}$ would increase $K(\mathcal{A})$ or dispatch entropy without positive frontier movement. □

Compression saturates when

$$\sup_m \mathrm{LCB}_t^{\mathrm{seq}}\, A(m) \le 0, \tag{136}$$

which occurs under non-reusable structure, unstable grounding, source churn, contradiction churn, high context entropy, high semantic entropy, high realization error, high $H(Y \mid M, C, \Gamma)$, nondiscriminating tests, non-composable tasks, verifier cost above value, active-dictionary entropy, branch explosion, maintenance/regression cost, failed equivalence witnesses, mode confusion, influence risk, or safety scope mismatch. The repair is not larger active memory alone; it is a better primitive, stronger test, narrower scope, context normalization, semantic factorization, source/rebuttal refresh, realization checking, hierarchical factorization, context-conditioned archival storage, lossy rate-distortion with explicit error, or abstention.

8 Training system and inefficiency controls

Training is execution-first. Before neural learning, implement the typed DSL/bytecode, context normalizer, semantic-contract compiler, source/evidence retriever, rebuttal retriever, entailment checker, round-trip realization checker, batch executor, verifier registry, falsifier, trace DAG, e-class store, provenance graph, authority lattice, declassification checker, influence-risk checker, cumulative-risk ledger, memory admission/demotion, and renderers. Neural modules are trained against traces produced by this executable system, not against unverifiable text alone.
Data are split by role:

$$\begin{aligned}
&\mathcal{D}_{all} = \mathcal{D}_{prop} \dot\cup \mathcal{D}_{ctx} \dot\cup \mathcal{D}_{sem} \dot\cup \mathcal{D}_{src} \dot\cup \mathcal{D}_{rebut} \dot\cup \mathcal{D}_{ent} \dot\cup \mathcal{D}_{real} \dot\cup \mathcal{D}_{dep} \dot\cup \mathcal{D}_{front} \dot\cup \mathcal{D}_{ver} \dot\cup \mathcal{D}_{cal} \dot\cup \mathcal{D}_{audit} \dot\cup \mathcal{D}_{replay} \dot\cup \mathcal{D}_{red},\\
&\mathcal{D}_{fit} = \mathcal{D}_{all} \setminus \mathcal{D}_{audit}.
\end{aligned} \tag{137}$$

Proposal models train on $\mathcal{D}_{prop}$; context normalizers on $\mathcal{D}_{ctx}$; semantic contractizers on $\mathcal{D}_{sem}$; source-recall estimators on $\mathcal{D}_{src}$; rebuttal-recall estimators on $\mathcal{D}_{rebut}$; entailment checkers on $\mathcal{D}_{ent}$; realization checkers on $\mathcal{D}_{real}$; dependence residualizers on $\mathcal{D}_{dep}$; frontier/admission predictors on $\mathcal{D}_{front}$; verifiers train on $\mathcal{D}_{ver}$; false-pass bounds are estimated on $\mathcal{D}_{cal}$; claims are reported only against $\mathcal{D}_{audit}$ or deployment replay; red-team attacks draw from $\mathcal{D}_{red}$. Cross-fitting rotates the roles, but certification is pairwise-disjoint at the event level. For $e = (\mu, a, v, T, r)$, set

$$\begin{aligned}
G_e &= \mathsf{split}_{gen}(\mu) \cup \mathsf{split}_{sel}(\mu) \cup \mathsf{split}_{ctx}(a) \cup \mathsf{split}_{sem}(a) \cup \mathsf{split}_{src}(a) \cup \mathsf{split}_{rebut}(a) \cup \mathsf{split}_{real}(a) \cup \mathsf{split}_{dep}(a) \cup \mathsf{split}_{front}(\mu),\\
W_e &= \mathsf{split}_{train}(v) \cup \mathsf{split}_{cal}(v) \cup \mathsf{split}_{audit},\\
A_e &= G_e \cup \mathsf{split}_{ent}(a) \cup \mathsf{split}_{train}(v) \cup \mathsf{split}_{cal}(v).
\end{aligned} \tag{138}$$

The certificate-carrying event must satisfy

$$G_e \cap W_e = \varnothing, \qquad \mathsf{split}_{train}(v) \cap \mathsf{split}_{cal}(v) = \varnothing, \qquad \mathsf{split}_{audit} \cap A_e = \varnothing. \tag{139}$$

The earlier three-way-intersection rule is insufficient: it allows a candidate and verifier to share data whenever calibration is separate. The same defect appears for context normalizers, semantic parsers, source/rebuttal retrievers, entailment checkers, and realization checkers. Here any leakage sets $\eta_{split} = 0$ and removes certificate weight; the trace can still be used for proposal learning or replay prioritization.
A trace is a DAG

$$\begin{aligned}
\tau = (N, E), \qquad n \in N &: (B_n, c_n, M_n, V_n, A_n, G_n, K_n, D_n, E_n^{sem}, C_n^{ctx}, R_n^{src}, R_n^{rebut}, \Omega_n^{real}, \mathrm{Dep}_n, \mathrm{Front}_n, \mathbf{r}_n),\\
e : n \to n' &= (a_e, o_e, cost_e, \mathbf{r}_e, \mathrm{Cert}_e, loss_e, ctx_e, sem_e, src_e, rebut_e, real_e, dep_e, front_e).
\end{aligned} \tag{140}$$

containing parses, context commitments, semantic atoms, source retrieval, rebuttal retrieval, entailment witnesses, realization witnesses, dependence blocks, vector-risk ledgers, frontier movements, invariants, sketches, programs, verifier choices, repairs, falsifier failures, active questions, support-loss bounds, gates, halts, memory writes, and final renderings. Teacher search defines a truncated posterior over certified traces

$$\begin{aligned}
p_*(\tau \mid T) \propto\ & \exp[-\mathcal{E}_*(\tau; T)]\mathbf{1}\{\mathrm{Cert}(\tau) \ge V_\Gamma\}\mathbf{1}\{\mathrm{Gate}(\tau) = 1\}\mathbf{1}\{\mathrm{Gate}_{\mathrm{NL}}(\mathrm{Render}(\tau)) = 1\}\\
&\cdot \mathbf{1}\{\epsilon_{support}(\tau) \le \epsilon_\Gamma\}\mathbf{1}\{\epsilon_{ctx}(\tau) \le \epsilon_{ctx,max}\}\mathbf{1}\{\epsilon_{sem}(\tau) \le \epsilon_{sem,max}\}\\
&\cdot \mathbf{1}\{\epsilon_{src}(\tau) \le \epsilon_{src,max}\}\mathbf{1}\{\epsilon_{rebut}(\tau) \le \epsilon_{rebut,max}\}\mathbf{1}\{\epsilon_{real}(\tau) \le \epsilon_{real,max}\}.
\end{aligned} \tag{141}$$

where $\mathcal{E}_*$ uses exact execution whenever available. Learned heuristics may order expansions but may not prune a trace unless they provide an admissible upper bound on lost certified utility. If no trace meets the threshold, the target is ask/abstain with calibrated uncertainty, not an answer.

The main objective is amortized free energy plus trace, value, calibration, verifier, semantic, source, entailment, safety, compression, support, and renderer losses:

$$\begin{aligned} L(\theta,\phi) = \mathbb{E}_{T\sim\mathcal{D}_{fit}}\Big[&\mathcal{J}_T(q_{\theta,\phi}) + \lambda_{base}L_{base} + \lambda_{cmp}L_{cmp} + \lambda_{tr}L_{trace} \\ &+ \lambda_{val}L_{value} + \lambda_{rep}L_{repair} + \lambda_{halt}L_{halt} + \lambda_{ver}L_{ver} + \lambda_{cal}L_{cal} \\ &+ \lambda_{safe}L_{safe} + \lambda_{mem}L_{mem} + \lambda_{supp}L_{supp} + \lambda_{ren}L_{render} \\ &+ \lambda_{ctx}L_{ctx} + \lambda_{sem}L_{sem} + \lambda_{src}L_{src} + \lambda_{rebut}L_{rebut} + \lambda_{ent}L_{ent} + \lambda_{real}L_{real} + \lambda_{tb}L_{TB} + \lambda_{mf}L_{mf} \\ &+ \lambda_{split}L_{split} + \lambda_{dom}L_{dom} + \lambda_{sub}L_{sub} + \lambda_{dep}L_{dep} + \lambda_{front}L_{front} + \lambda_{probe}L_{probe}\Big], \end{aligned} \tag{142}$$

$$\begin{aligned} \mathcal{J}_T(q) = &\mathbb{E}_{q(\mu,c,\mathfrak{n},V)}[\mathcal{E}(\mu,c,\mathfrak{n},V;T,\Gamma)] \\ &+ \mathrm{KL}(q(\mu,c,\mathfrak{n},V \mid T)\|p_0(\mu,c,\mathfrak{n},V \mid \Gamma)). \end{aligned} \tag{143}$$

The substrate loss L_{base} is defined on audited observation, predictive-state, memory-recall, context, semantic, source, rebuttal, entailment, realization, projection, and routing targets above. Substrate calibration is separately penalized by

$$L_{sub} = \mathrm{ECE}(\hat{\epsilon}_{sub}) + \lambda_{miss}\mathrm{BCE}(\widehat{\mathsf{miss}}_\theta, \mathsf{miss}) + \lambda_{bound}\max\{0, \widehat{\Pr}(\mathsf{miss}) - \epsilon_{sub}\}. \tag{144}$$

Dependence, frontier, and critical-edge losses remove three training inefficiencies: duplicated correlated evidence, macros that improve training likelihood but not held-out compression, and neural states that omit the sparse edges later needed for certification:

$$L_{dep} = \mathrm{BCE}(\widehat{\mathrm{Leak}}_{dep}, \mathrm{Leak}_{dep}) + \mathrm{ECE}(\hat{\epsilon}_{dep}) + \lambda_{res}\|\widehat{X}_g^\perp - X_g^\perp\|_1 + \lambda_{bound}\max\{0, \widehat{\Pr}(\mathrm{Leak}_{dep}) - \epsilon_{dep}\}, \tag{145}$$

$$L_{front} = \|\widehat{\Delta\,\mathrm{Front}}_\theta - \Delta\,\mathrm{Front}\|_2^2 + \mathrm{BCE}(\widehat{\mathrm{Admit}}, \mathbf{1}\{\Delta\,\mathrm{Front} > 0\}) + \lambda_{dup}\widehat{\Delta I} + \lambda_{rd}\max\{0, \hat{d}_\Gamma - D\}, \tag{146}$$

$$L_{probe} = \mathrm{BCE}(\widehat{\mathrm{keep}}_{crit}, \mathbf{1}\{e \in \mathrm{Crit}(T)\}) + \lambda_{swap}\mathrm{CE}(\widehat{S}_\Gamma(\mathrm{swap}(e)), S_\Gamma(\mathrm{swap}(e))) + \lambda_{rehyd}\max\{0, \epsilon_{crit} - \epsilon_{crit,max}\}. \tag{147}$$

Compiler learning marginalizes over parse alternatives rather than selecting a single parse:

$$L_{cmp} = -\log\sum_{k=1}^{K} q_\phi(c_k, \mathfrak{n}_k, \Gamma_k, A_k, P_k \mid O, H)\mathbf{1}\{c_k \in \mathcal{C}^*(T), \mathfrak{n}_k \in \mathcal{N}^*(T)\} + \lambda_\Sigma\mathrm{NLL}_{cal}(\Sigma_k) + \lambda_b b_k + \lambda_{bctx}b_k^{ctx} + \lambda_{bsem}b_k^{sem} + \lambda_{breal}b_k^{real}. \tag{148}$$

Context, semantic, source, rebuttal, entailment, and realization losses are optimized as recall-first gates rather than fluent-text objectives:

$$L_{ctx} = -\log\sum_{\mathfrak{n}\in\mathcal{N}^*(T)} q_\psi(\mathfrak{n} \mid U, H, \Gamma) + \lambda_{ref}\mathrm{CE}(\widehat{D}, D) + \lambda_{gran}\mathrm{CE}(\widehat{\gamma}, \gamma) + \lambda_{ask}\mathrm{BCE}(\widehat{\mathrm{Ask}}, \mathbf{1}\{\mathrm{CtxOK} = 0\}), \tag{149}$$

$$L_{sem} = -\log\sum_{\mathfrak{n}\in\mathcal{N}^*(T)} q_\psi(\mathfrak{n} \mid U, H, \Gamma) + \lambda_{mode}\mathrm{CE}(\widehat{m}, m) + \lambda_{dep}\mathrm{CE}(\widehat{\Delta}, \Delta) + \lambda_{ask}\mathrm{BCE}(\widehat{\mathrm{Ask}}, \mathbf{1}\{\mathrm{SemOK} = 0\}), \tag{150}$$

$$L_{src} = \mathrm{BCE}(\widehat{\mathsf{miss}}_{src}, \mathsf{miss}_{src}) + \mathrm{ECE}(\hat{\epsilon}_{src}) + \lambda_{bound}\max\{0, \widehat{\Pr}(\mathsf{miss}_{src}) - \epsilon_{src}\}, \tag{151}$$

$$L_{rebut} = \mathrm{BCE}(\widehat{\mathsf{miss}}_{rebut}, \mathsf{miss}_{rebut}) + \mathrm{ECE}(\hat{\epsilon}_{rebut}) + \lambda_{bound}\max\{0, \widehat{\Pr}(\mathsf{miss}_{rebut}) - \epsilon_{rebut}\}, \tag{152}$$

$$L_{ent} = \mathrm{BCE}(\widehat{\mathrm{Ent}}(P_a, e_a, \Omega_a), \mathrm{Ent}^*) + \lambda_{hard}\max\{0, 1 + s_\theta(a^-) - s_\theta(a^+)\}, \tag{153}$$

$$L_{real} = \mathrm{BCE}(\widehat{\mathrm{Hidden}}_{cert}, \mathrm{Hidden}_{cert}) + \mathrm{ECE}(\hat{\epsilon}_{real}) + \lambda_{rt}\mathrm{CE}(\widehat{\mathrm{Atom}}(y), \mathrm{Atom}(y)) + \lambda_{bound}\max\{0, \widehat{\Pr}(\neg\,\mathrm{RealOK}) - \epsilon_{real}\}. \tag{154}$$

Trace imitation is advantage-weighted and off-policy corrected, but only when effective sample size is adequate:

$$\bar{w}_e = \mathrm{clip}\left(\frac{p_*(\tau \mid T)}{q_{old}(\tau \mid T)}, 0, w_{max}\right)\max\{0, A_e\}, \qquad \mathrm{ESS} = \frac{(\sum_e \bar{w}_e)^2}{\sum_e \bar{w}_e^2}, \tag{155}$$

$$L_{trace} = \mathbf{1}\{\mathrm{ESS} \geq N_{ess}\}\left[-\sum_{e\in\tau_*}\bar{w}_e\log\pi_\theta(a_e \mid B_e, \Gamma)\right]. \tag{156}$$

If $\mathrm{ESS} < N_{ess}$, the batch is used for replay prioritization or new teacher search, not for high-variance imitation.

The value and halting heads learn the expected marginal certified utility of another step:

$$\begin{aligned} L_{value} &= \sum_r \|v_\theta(B^{(r)}) - \mathrm{EVC}_r^{MC}\|_2^2, \\ L_{halt} &= \sum_r \mathrm{BCE}(\sigma(g_\theta(B^{(r)})), \mathbf{1}\{\mathrm{EVC}_r \leq 0 \vee \mathrm{Cert}_r \geq V\}). \end{aligned} \tag{157}$$

Repairs and near-miss negatives train discrimination at the decision boundary:

$$L_{repair} = -\log p_\theta(\rho^* \mid B, \mu^-, x^*) + \lambda_{rank}\max\{0, 1 + E_\theta(\mu^+) - E_\theta(\mu^-)\}. \tag{158}$$

Verifier training calibrates false-pass probability, not merely pass accuracy:

$$L_{ver} = \sum_v [-f_v\log\hat{p}_v - (n_v - f_v)\log(1 - \hat{p}_v)] + \lambda_{adv}\mathbb{E}_{x^*}\ell(v(\mu, x^*), S_\Gamma(\mu, x^*)). \tag{159}$$

Calibration uses binned, distributional, and anytime penalties:

$$L_{cal} = \mathrm{ECE}(q) + \mathbb{E}[(\Pr_q[S_\Gamma = 1 \mid B] - \mathbf{1}\{S_\Gamma = 1\})^2] + \sum_v \mathrm{ECE}(\hat{p}_v) + \lambda_{cs}\sum_{v,r}\max\{0, \hat{p}_{v,r} - p_{v,r}^+\}. \tag{160}$$

Split hygiene and multi-fidelity verification are trained as predictors but enforced as hard constraints:

$$L_{split} = \sum_{e} \mathrm{BCE}(\widehat{\mathrm{leak}}_\theta(e), \mathbf{1}\{\Pi_{\mathrm{split}}(e) = 0\}) + \lambda_{audit}\mathbf{1}\{\mathrm{Sel}(e) \cap \mathcal{D}_{audit} \neq \varnothing\}, \tag{161}$$

$$L_{mf} = \sum_{j<k} \mathrm{CE}(\hat{r}_{j\to k}, \mathbf{1}\{\text{stage } k \text{ changed certificate or risk}\}). \tag{162}$$

Cheap verifier stages may reject or route candidates. They may certify only through their own calibrated e-values; otherwise certificate weight is assigned to the terminal independent stage. Safety is trained as prediction but enforced as a hard gate:

$$\begin{aligned} L_{safe} = \mathrm{CE}(\hat{\alpha}_\theta, \alpha) + \|\widehat{\Delta\mathbf{r}}_\theta - \Delta\mathbf{r}\|_1 + \lambda_{cum}\|\widehat{\mathbf{r}}_{cum} - \mathbf{r}_{cum}\|_1 + \lambda_{infl}\mathrm{CE}(\widehat{\mathrm{Infl}}, \mathrm{Infl}) + \lambda_{inj}\ell_{inj} + \lambda_{conf}\ell_{conflict}, \\ \text{Gate is not relaxed during execution.} \end{aligned} \tag{163}$$

Support learning predicts lost mass and trains recall before precision:

$$\begin{aligned} L_{supp} = \sum_r \Big[&\max\{0, \epsilon^{(r)}_{prune} - \epsilon_{max}\} + \mathrm{BCE}(\hat{m}^{(r)}_i, \mathbf{1}\{(j,k) \in \mathcal{N}^{oracle}_i\}) \\ &+ \lambda_{ctx}\mathrm{BCE}(\widehat{\mathrm{keep}}_{ctx}, \mathbf{1}\{C^{ctx} \in C^{oracle}\}) + \lambda_{sem}\mathrm{BCE}(\widehat{\mathrm{keep}}_{sem}, \mathbf{1}\{a \in \mathcal{A}^{oracle}\}) + \lambda_{src}\mathrm{BCE}(\widehat{\mathrm{keep}}_{src}, \mathbf{1}\{S \subset R^{oracle}\}) \\ &+ \lambda_{rebut}\mathrm{BCE}(\widehat{\mathrm{keep}}_{rebut}, \mathbf{1}\{S^- \subset R^{oracle}_-\}) + \lambda_{real}\mathrm{BCE}(\widehat{\mathrm{keep}}_{real}, \mathbf{1}\{\Omega^{real} \in \Omega^{oracle}\})\Big]. \end{aligned} \tag{164}$$

Compression learning predicts admission gain while the actual decision remains the confidence-bound rule:

$$\begin{aligned} L_{mem} =& \|\hat{A}_\theta(m) - A_t(m)\|_2^2 + \mathrm{BCE}(\hat{d}_\theta(m), \mathbf{1}\{\mathrm{LCB}\, A_t(m) > 0\}) + \lambda_{scope}\mathrm{CE}(\hat{\omega}_m, \omega_m) \\ &+ \lambda_{ctx}\mathrm{CE}(\hat{\rho}^{ctx}_m, \rho^{ctx}_m) + \lambda_{sem}\mathrm{CE}(\hat{\sigma}^{sem}_m, \sigma^{sem}_m) + \lambda_{src}\mathrm{CE}(\hat{\rho}^{src}_m, \rho^{src}_m) + \lambda_{rebut}\mathrm{CE}(\hat{\rho}^{rebut}_m, \rho^{rebut}_m) + \lambda_{real}\mathrm{CE}(\hat{\rho}^{real}_m, \rho^{real}_m) \\ &+ \lambda_{dep}\mathrm{ECE}(\hat{\epsilon}_{dep,m}) + \lambda_{front}\|\widehat{\Delta\,\mathrm{Front}}_m - \Delta\,\mathrm{Front}_m\|_2^2 + \lambda_R\|\widehat{\Delta\mathbf{r}}_m - \Delta\mathbf{r}_m\|_1. \end{aligned} \tag{165}$$

For generative mechanism proposals, a GFlowNet trajectory-balance loss samples diverse certified mechanisms:

$$\begin{aligned} R(\mu, T) =& \epsilon + \exp[-\mathcal{E}(\mu; T)]\mathbf{1}\{\mathrm{Cert}(\mu, T) \geq V_\Gamma\}\mathbf{1}\{\mathrm{Gate}(\mu) = 1\}\mathbf{1}\{\epsilon_{support}(\mu) \leq \epsilon_\Gamma\} \\ &\cdot \mathbf{1}\{\epsilon_{ctx}(\mu) \leq \epsilon_{ctx,max}\}\mathbf{1}\{\epsilon_{sem}(\mu) \leq \epsilon_{sem,max}\}\mathbf{1}\{\epsilon_{src}(\mu) \leq \epsilon_{src,max}\}\mathbf{1}\{\epsilon_{rebut}(\mu) \leq \epsilon_{rebut,max}\}\mathbf{1}\{\epsilon_{real}(\mu) \leq \epsilon_{real,max}\} \\ &\cdot \mathbf{1}\{\epsilon_{dep}(\mu) \leq \epsilon_{dep,max}\}\mathbf{1}\{\Delta\,\mathrm{Front}(\mu) > 0\}\mathbf{1}\{\Delta\mathbf{r}(\mu) \preceq \mathbf{B}_R - \mathbf{r}_t\}. \end{aligned} \tag{166}$$

$$L_{TB} = \left(\log Z_\theta(T) + \sum_r \log P_F(a_r \mid s_r, T) - \log R(\mu, T) - \sum_r \log P_B(a_r \mid s_{r+1}, T)\right)^2. \tag{167}$$

Renderer training is conditioned on certified atoms and cannot invent unsupported claims:

$$\begin{aligned} L_{render} =& -\log p_\theta(y_{cert} \mid Z_{cert}, \mathrm{Atom}(y), \Gamma) + \lambda_{unc}\mathrm{CE}(\hat{E}(y), E(y)) + \lambda_{ctx}\mathrm{CE}(\hat{C}(y), C^{ctx}(y)) + \lambda_{src}\mathrm{CE}(\hat{P}(y), P(y)) \\ &+ \lambda_{rebut}\mathrm{CE}(\hat{R}^-(y), R^-(y)) + \lambda_{real}\mathrm{BCE}(\widehat{\mathrm{RealOK}}(y), \mathrm{RealOK}(y)) + \lambda_{atom}\mathrm{CE}(\widehat{\mathrm{Atom}}(y), \mathrm{Atom}(y)) + \lambda_{hall}\mathbf{1}\{\mathrm{Def}(y) \neq \varnothing\}. \end{aligned} \tag{168}$$

Loss weights are normalized by gradient scale so cheap text or imitation losses cannot overwhelm verification, calibration, context recall, semantic recall, source recall, rebuttal recall, realization recall, support recall, or safety:

$$\lambda_i \leftarrow \mathrm{clip}_{[l_i, u_i]}\left\{\lambda_i \left(\frac{\mathrm{median}_j \bar{g}_j}{\bar{g}_i + \epsilon}\right)^\alpha\right\}, \qquad \bar{g}_i \leftarrow (1 - \beta_g)\bar{g}_i + \beta_g\|\nabla_\theta L_i\|_2. \tag{169}$$

Training is block-coordinate with frozen audit labels and hard gates; otherwise cheap language likelihood would create gradient paths that appear to improve utility while eroding certification. Let

$$\Theta = (\psi_{ctx}, \psi_{sem}, \rho_{src}, \rho_{rebut}, \rho_{ent}, \rho_{real}, \rho_{dep}, \xi_{ver}, \theta_{sub}, \phi_{cmp}, \omega_{ren}, \kappa_{mem}, \kappa_{front}). \tag{170}$$

Then

$$(\psi^+_{ctx}, \psi^+_{sem}) = \arg\min_{\psi} L_{ctx} + L_{sem} + \lambda_{ask}L_{ask}, \tag{171}$$

$$(\rho^+_{src}, \rho^+_{rebut}, \rho^+_{ent}, \rho^+_{real}, \rho^+_{dep}, \xi^+_{ver}) = \arg\min_{\rho,\xi} L_{src} + L_{rebut} + L_{ent} + L_{real} + L_{dep} + L_{ver} + L_{cal} + L_{split}, \tag{172}$$

$$(\theta^+_{sub}, \phi^+_{cmp}) = \arg\min_{\theta,\phi} \mathcal{J}_T + L_{base} + L_{cmp} + L_{trace} + L_{value} + L_{halt} + L_{supp} + L_{probe}, \tag{173}$$

$$(\omega^+_{ren}, \kappa^+_{mem}, \kappa^+_{front}) = \arg\min_{\omega,\kappa} L_{render} + L_{mem} + L_{safe} + L_{front}, \tag{174}$$

where certificate, split, dependence, context, semantic, source, rebuttal, realization, frontier, and vector-safety gates are evaluated as detached constraints during training and exactly during execution.

Training compute is allocated by measured marginal certified utility and by dual prices for scarce exact execution, dependence calibration, risk calibration, and replay. The optimizer is over budget quanta, not over static epoch counts:

$$\begin{aligned} \mathbf{B}^* = \arg\max_{\sum_i B_i \leq B} \ & \widehat{S}_{\mathrm{ver}}(\mathbf{B}) - \lambda_{overfit}\widehat{\mathrm{Gap}} - \lambda_{unsafe}\widehat{\mathrm{GateFail}} - \lambda_{leak}\widehat{\mathrm{SplitLeak}} \\ &- \lambda_{supp}\widehat{\epsilon}_{support} - \lambda_{sub}\widehat{\epsilon}_{sub} - \lambda_{dep}\widehat{\epsilon}_{dep} - \lambda_{ctx}\widehat{\epsilon}_{ctx} - \lambda_{sem}\widehat{\epsilon}_{sem} - \lambda_{src}\widehat{\epsilon}_{src} \\ &- \lambda_{rebut}\widehat{\epsilon}_{rebut} - \lambda_{real}\widehat{\epsilon}_{real} - \boldsymbol{\lambda}_R^\top\widehat{\mathbf{r}}_{cum} - \lambda_{front}\widehat{\Delta\,\mathrm{Front}}^- - \lambda_{out}\widehat{\mathrm{OutDom}} - \lambda_{idle}\mathrm{IdleExact}, \end{aligned} \tag{175}$$

$$\mathbf{B} = (B_{sub}, B_{model}, B_{data}, B_{ctx}, B_{sem}, B_{src}, B_{rebut}, B_{ent}, B_{real}, B_{dep}, B_{front}, B_{search}, B_{replay}, B_{ver}, B_{red}, B_{exact}, B_{risk}). \tag{176}$$

At an interior optimum the observed marginal returns satisfy the KKT balancing condition

$$\frac{\partial\widehat{S}_{\mathrm{ver}}}{\partial B_i} - \lambda^{loss}_i\frac{\partial\widehat{\epsilon}_i}{\partial B_i} - \boldsymbol{\lambda}_R^\top\frac{\partial\widehat{\mathbf{r}}}{\partial B_i} - \lambda_{front}\frac{\partial\widehat{\Delta\,\mathrm{Front}}^-}{\partial B_i} = \lambda_0, \tag{177}$$

with boundary modules receiving budget only when their estimated left derivative is positive. This prevents cheap imitation, renderer likelihood, or oversized neural pretraining from consuming exact execution and calibration budget while the active certificate bottleneck remains elsewhere. Teacher search is spent on posterior-boundary tasks, not uniformly:

$$e^* = \arg\max_e \frac{H(M, C^{ctx}, D^{sem}, R^{src}, R^{rebut}, \Omega^{real}, \mathrm{Dep}, \mathrm{Front} \mid D) - \mathbb{E}_o H(M, C^{ctx}, D^{sem}, R^{src}, R^{rebut}, \Omega^{real}, \mathrm{Dep}, \mathrm{Front} \mid D \cup \{(e,o)\}) + \mathbb{E}\Delta S_{\mathrm{ver}} + \lambda_{reg} \Pr(\mathrm{regression} \mid e)}{c(e) + c_{label}(e) + c_{exec}(e) + c_{ctx}(e) + c_{sem}(e) + c_{src}(e) + c_{rebut}(e) + c_{real}(e) + c_{dep}(e) + c_{front}(e) + \boldsymbol{\lambda}_R^\top \mathbf{r}(e)}. \tag{178}$$

Efficiency constraints are structural. All modules read and write a content-addressed trace factor graph, so a parse, source set, rebuttal set, entailment witness, verifier result, realization check, dependence block, frontier measurement, or risk estimate is computed once per hash and reused by every head whose split policy permits it. Candidate execution is batched by operator signature and e-class; context and semantic parses are cached by discourse hash, time scope, reference class, and ambiguity class; source and rebuttal retrieval are cached by atom, scope, freshness, and split; realization checks are cached by atom-plan hash and renderer; dependence residuals are cached by block identity; entailment/verifier calls are batched and cached by $(v, \mu, a, x, scope, split)$; partial executions are memoized as $(state, lower\ bound, upper\ bound, prune\ reason)$; failed verifications create negative certificates for siblings in the e-graph when sound; replay is prioritized by surprise, regression, staleness, support loss, context loss, semantic loss, source staleness, rebuttal staleness, realization drift, dependence leakage, frontier regression, and capability risk; and exact stages are invoked on the uncertainty boundary where their expected certificate or recall gain exceeds cost. External LLM proposals may be used only as ablated teachers with logged budget and no certificate authority; an unlogged external model is hidden compute. At inference, VNS_θ may render style, paraphrase, and nonclaim glue, but every factual, mathematical, executable, advice-like, private, or side-effecting token must trace to certified context, atoms, parents, source/rebuttal records, entailment witnesses, and realization witnesses.

The curriculum has gates. C_0: deterministic grids, strings, FSMs, small graphs, arithmetic, finite-state tasks, and data transformations with exact verifiers. C_1: hidden-schema splits, equality saturation, typed program synthesis, and theorem-proving fragments. C_2: noisy/partial observations, active tests, small causal worlds, and verifiable planning. C_3: formal tools, agent workflows, tool use, authority, declassification, prompt-injection, rollback, influence-risk, and conflict. C_4: controlled dialogue/artifacts with context normal forms, semantic atom extraction, source-grounded QA, contradiction search, entailment checking, round-trip realization checking, calibrated uncertainty, and intent-entropy gates. C_5: continual compression with replay-safe macro admission. Advancement requires held-out certified utility, calibrated anytime false-pass bounds, calibrated dependence residuals, positive frontier movement, sublinear active-memory growth, bounded support, context, semantic, source, rebuttal, realization, dependency, and substrate loss, vector-risk budget validity, and zero critical gate violations in adversarial suites.

9 Evaluation, failure, and minimal implementation

Report certified utility, solve rate at threshold, answer accuracy stratified by evidence level, domain-route confusion, intent-entropy failures, context/reference recall, vague-predicate definition failures, semantic-parse recall, mode confusion, entailment false-support rate, source-recall miss rate, rebuttal-recall miss rate, dependence-leakage rate, round-trip realization loss, citation/context loss, calibrated uncertainty quality, ECE, anytime verifier false-pass/false-fail bounds, counterexample discovery, support loss, critical-edge omission loss, substrate omission loss, vector-risk consumption by channel, influence-risk consumption, active-memory growth, archival growth, rate-distortion frontier movement, transfer lift, dual-allocation regret, wall-clock/FLOPs, hidden search, human bits, safety-gate violations, privacy/declassification failures, macro admissions/demotions, branch factor after compression, interference entropy, and regression after sleep compression. Required ablations remove VNS_θ typed slots, selective recurrence, discourse state, context normalizer, semantic atoms, source-recall guard, rebuttal-recall guard, realization guard, dependence residualization, vector-risk gate, frontier-valid compression, critical-edge probes, entailment gate, executable projection, parse posterior, domain routing, support-loss guard, substrate-recall guard, type masks, authority masks, cumulative-risk budget, influence-risk gate, e-graph merging, verifier feedback, falsifier, active tests, residual escrow, macro demotion, cross-fitting, loss balancing, replay, dual compute allocation, and renderer gates.

Criterion 1 (Architectural failure)**.** *VPM-5.3 fails as a base architecture if it cannot beat enumerative search and same-budget transformer, state-space, and neural program-synthesis baselines on hidden-schema mechanism tasks; if* VNS_θ *omits certifying traces at nonvanishing rate after exact rehydration; if context normalization collapses unresolved references or vague predicates into a single unsupported parse; if semantic contractization collapses ambiguous user language into a single unsupported parse; if source retrieval misses minimally sufficient evidence sets at nonvanishing rate; if rebuttal retrieval misses material defeaters at nonvanishing rate; if round-trip realization permits hidden factual atoms; if entailment checking permits unsupported paraphrases; if active memory grows linearly with solved tasks; if compiler, sparse-state, context, semantic, source, rebuttal, realization, or substrate support loss dominates after active tests; if near-miss attacks keep verifier false-pass bounds high; if dependence residuals or split leakage create certificate inflation; if gains require hidden test-time compute; if residuals carry most capability without certificates; if compression requires opaque macros without equivalence witnesses or positive held-out frontier movement; if cumulative risk is hidden in many small gated actions or by scalar cancellation; if*

influence risk is hidden in soft spans; if safety gates are bypassed by data-only inputs; if verifiers certify same-split generators; if support-recall or dependence calibration is miscalibrated under shift; if broad language/world-knowledge claims are rendered when $\mathrm{DomOK} = 0$ *or* $\mathrm{Gate}_{\mathrm{NL}} = 0$*; or if inference depends on an external LLM as an uncertified cognitive component.*

A minimal VPM-0 implements VNS_θ with typed event encoders, selective recurrence, slots, discourse state, context normal forms, semantic atoms, source recall, rebuttal recall, entailment checking, round-trip realization checking, executable projection, and calibrated substrate recall on typed grids, strings, finite-state machines, small graphs/mazes, simple diagrams, data transformations, theorem fragments, source-grounded QA over a fixed corpus, controlled dialogue, and small causal worlds; exact primitive verifiers; dependence-block residualization; falsifier-generated near misses; active questions; an authority lattice; vector-risk ledgers; e-value certificates; support-loss bounds; frontier-valid macro admission/demotion; and matched transformer/state-space/enumerative baselines. The primary success signal is a compression phase transition: after replay-safe macro admission, solved-task cost falls, active memory grows sublinearly, held-out certified utility rises, rate-distortion frontier decreases, support, context, semantic, source, rebuttal, realization, dependency, and substrate loss remain bounded, vector-risk budgets remain valid, and false-pass bounds do not rise.

A Compact inference procedure

```
VPM-Infer(O, A, Gamma, B):
    H <- VNS_EncodeUpdate(O, Gamma) # typed predictive-state substrate
    Psi <- ConversationNormalFormPosterior(O, H, Gamma) # context, discourse, atoms, modes, ambiguity
    if not CtxOK(Psi) or not SemOK(Psi): return AskOrScopedUncertainty(Psi, Gamma)
    Gamma <- RefineContract(Gamma, Psi)
    Xi <- EstimateCertifiability(H, Psi, Gamma)
    Beta <- BottleneckVector(H, Psi, Gamma, Xi)
    route <- DomainRoute(Xi, Gamma)
    if route == ground:
        Rsrc <- RetrieveSources(Psi.atoms, Gamma)
        Rreb <- RetrieveRebuttals(Psi.atoms, Gamma)
        Xi <- UpdateSourceRebuttalBounds(Xi, Rsrc, Rreb)
    if route in {Ask, Abstain, archive}: return RouteResponse(route, H, Gamma)
    if route == narrow: O, Gamma <- NarrowScopeOrExactRehydrate(O, H, Psi, Gamma)
    C <- CompilerPosterior(O, Psi, Gamma, H) # alternatives, provenance, authority
    R <- RetrieveActive(C, Gamma, H) # only admitted capsules; archive is proposal data
    S <- InitSlots(C, R, Psi, H, Gamma)
    Alpha <- AllocateClaimFamilyBudget(Gamma, Psi.atoms)
    for r in 0..Rmax:
        H <- VNS_EncodeUpdate(O, Gamma, S)
        S <- MechanismStateCell_theta(S, C, R, H, Gamma)
        eps, eps_sub, eps_ctx, eps_sem, eps_src, eps_rebut, eps_real, eps_dep, rvec, dfront <- CalibratedLosses(S, H, Psi,
             exact_recall_audit)
        if eps > Gamma.epsilon or eps_sub > Gamma.epsilon_sub or eps_dep > Gamma.epsilon_dep or eps_ctx > Gamma.epsilon_ctx
             or eps_sem > Gamma.epsilon_sem or eps_src > Gamma.epsilon_src or eps_rebut > Gamma.epsilon_rebut or eps_real >
             Gamma.epsilon_real or not RiskBudgetOK(rvec):
            S <- RehydrateOrWidenOrGroundOrAsk(S, Psi, Beta)
            route <- DomainRoute(EstimateCertifiability(H, Psi, Gamma), Gamma)
            if route in {Ask, Abstain, narrow}: return RouteResponse(route, H, Gamma)
        sketches <- ProjectInvariantsAndEClasses(S)
        programs <- EscalateIfEVCPositive(sketches, S, eps, eps_ctx, eps_sem, eps_src, eps_rebut, eps_real)
        programs <- ExpandMacrosWithWitnesses(programs)
        routes <- MultiFidelityRoute(programs, S, Gamma)
        routes <- DependencyBlockAndRiskRoute(routes, S, Gamma)
        results <- BatchExecuteVerifyEntailRebutRealize(routes, C, Psi, Gamma, Alpha) # block e-value ledger
        attacks <- FalsifyNearMisses(programs, results, Gamma)
        S <- InjectEvidence(S, results, attacks, eps, eps_ctx, eps_sem, eps_src, eps_rebut, eps_real, eps_dep, rvec, dfront)
        y <- CandidateRender(S, Psi, Gamma)
        if ContractSatisfied(S, Gamma) and RenderGate(y, S, Gamma) and Gate_NL(y, S, Gamma) and RealOK(y, PlannedAtoms(S), S
             , Gamma):
            return RenderCertified(y, S, Gamma)
        if ClarificationDominates(S, Gamma):
            return RenderQuestion(S, Gamma)
        a <- ProposedAction(S)
        if a and Gate(a, S, Gamma) and VectorRiskOK(a, S, Gamma):
            O <- SafeToolOrEnvStep(a); C,R,S <- Update(O,C,R,S)
        if EVC(S) <= 0:
            break
    return BestEffortWithEvidenceLevelOrRefusal(S, Gamma)
```

B Executable sanity conditions

The implementation test kernel must check these behavior-level invariants: domain routing solves cheap-verifier reusable tasks and abstains/narrows when $p_v^+ = 1$, $\eta_v = 0$, unresolved dependence, high shift, high intent entropy, high context entropy, high semantic entropy, unit source loss, unit rebuttal loss, unit realization loss, vector-risk exhaustion, or unit support/substrate loss; correlated verifiers do not multiply evidence unless residualized by dependence blocks; adaptive claim-family spending is finite; same-split verifier/candidate/context/semantic/source/rebuttal/realization/dependence/frontier events have zero certificate weight; pairwise split leakage is rejected rather than merely triple-overlap leakage; support, dependency, context, semantic, source, rebuttal, realization, critical-edge, and substrate caps are nonnegative and equal zero at unit loss; componentwise vector risk blocks actions even when a scalar average is small; data-authority tool output cannot satisfy instruction-authority gates; explicit grants do not bypass permission, certificate, impact, budget, privacy, sandboxing, cumulative-risk, influence-risk, dependency-risk, or conflict conjuncts; residuals cannot certify side-effecting actions; ambiguity-increasing, source-stale, rebuttal-stale, realization-drifting, dependence-leaking, frontier-negative, risk-increasing, or branch-increasing macros are rejected by sequence-valid admission; active macros have expansion/equivalence witnesses and positive held-out frontier movement; LCB admission and UCB demotion are symmetric; EVC halts when marginal verified utility is below cost; compiler canonicalization is reversible or support-bounded; VNS_θ retrieval and critical-edge miss rates are calibrated against exact teacher traces; source retrieval miss rates are calibrated against sufficient evidence sets; rebuttal retrieval miss rates are calibrated against material defeaters; realization gates reject hidden certified atoms; entailment gates reject unsupported paraphrase; falsifiers reject training-set coincidences; dual compute allocation gives no budget to heads whose marginal certified return is negative; and renderer claims inherit context, source, rebuttal, realization, evidence, privacy, family budget, support, substrate, semantic, dependence, vector-risk, domain, and authority labels.

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